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Sequence variants affect the genome-wide rate of germline microsatellite mutations

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Sequence variants affect the genome-wide rate of germline microsatellite mutations

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Abstract

Microsatellites are polymorphic tracts of short tandem repeats (STRs) with one to six base-pair (bp) motifs and are some of the most polymorphic markers in the genome. Using 6,084 Icelandic parent-offspring trios we estimate 63.7 (95% CI: 61.9-65.4) microsatellite de novo mutations (mDNMs) per offspring per generation. Paternal mDNMs occur at longer repeats, while maternal mDNMs affect more bp. mDNMs increase by 0.97 (95% CI: 0.90–1.04) and 0.31 (95% CI: 0.25-0.37) per year of father's and mother's age at conception, respectively. We found two independent coding variants associated with an increased number of mDNMs transmitted to offspring; A missense variant (allele frequency (AF) = 1.9%) in *MSH2*, a mismatch repair gene, increases transmitted mDNMs from both parents (effect: 13.1 paternal and 7.8 maternal mDNMs). A synonymous variant (AF = 20.3%) in *NEIL2*, a DNA damage repair gene, increases paternally transmitted mDNMs (effect: 4.4 mDNMs). Thus, the microsatellite mutation rate in humans is in part under genetic control.

Introduction

Mutations enable life to evolve and adapt. Accurate estimates of the rate of mutations and the processes behind them are therefore imperative for understanding evolution, making inferences about population history[1–6] and understanding the genetics of disease and other phenotypes[7–11].

Around 3% of the human genome are short tandem repeats (STRs)[12], some of which are polymorphic, i.e. microsatellites, and mutate several orders of magnitude faster than non-repetitive sequences[13]. The forces that shape age related mDNM accumulation in the two sexes and the effects of parental genotypes on accumulation is largely unknown.

Genetic factors responsible for genome integrity can be expected to play a role in DNM accumulation. Sequence variants that increase single nucleotide polymorphism (sDNM) and mDNM rates are known in somatic tissues in humans [14] and sequence variants that increase the sDNM rate are known in animal and yeast models[15,16]. Apart from a handful of clinical cases[17,18],

no human germline mutators are known that affect the de novo mutation rate of microsatellites or other types of variants.

Most mDNMs are believed to occur as a result of failure in the processes responsible for sequence fidelity during DNA replication. These processes verify the correct pairing of bp and replace incorrectly paired or damaged DNA bases[19,20]. Loss of function mutations affecting genes responsible for sequence fidelity are known to cause somatic microsatellite instability, which in turn can result in increased risk of colorectal, gastric, endometrial and other types of cancer[21].

Recombination, DNA damage repair and nonhomologous end joining (NHEJ) have also been implicated as determinants of mDNMs[22,23] and since the two germlines have different exposure to these processes it is logical to assume that the mDNM accumulation is different between the sexes. Spermatogonia undergo mitosis continuously, increasing the risk of replication errors while the recombination rate is higher in females[24] and oocytes wait in dictyate arrest from the mother's birth to reproduction, exposed to DNA damage to be repaired by NHEJ or homologous recombination[25]. The rate of mDNMs in humans has been shown to vary by a number of microsatellite properties [26–28], parent of origin, and paternal[29] age, but researchers have to date been unable to detect a relationship with maternal age.

Previous studies of mDNMs have largely been confined to small sets of well-characterized microsatellites or focused on specific diseases [1,29]. Here, we identified and genotyped microsatellites in two large sequencing cohorts with the aim of estimating polymorphism and mDNM rate and identifying environmental and genetic determinants of mDNMs.

Results

STR identification and genotyping

We used popSTR[30] to call genotypes at 5,401,401 autosomal short tandem repeats (STRs) in two large WGS datasets, 53,026 Icelanders and 150,119 participants in the UKB, sequenced to

an average coverage of 39.2x (min: 19.7, max: 608.3) and 32.5x (min: 23.6, max: 128.1), respectively.

The STRs were identified by Tandem repeats finder[31,32]. Each STR has a repeat motif along with start and end positions in the reference and we refer to the sequence between these positions as the reference repeat tract (RRT) and its length as RRT length. For motif lengths between one and three bp, we compared the behavior of motif equivalence classes where all members in a class are either a circular shift or a reverse complement of each other. For example, the members of the AAT motif equivalence class are: AAT, ATA, TAA, ATT, TAT and TTA (Table S1).

We found 1,394,292 (25.8%) and 2,393,292 (44.3%) of the STRs to be polymorphic in the Icelandic and UK datasets, respectively and will refer to these polymorphic STRs as microsatellites. We describe microsatellite diversity through polymorphism rate (the fraction of STRs that are polymorphic) and expected heterozygosity[32,33] (Fig. S*1*, Fig. S*2*, Table S18, Table S19, Table S20, Table S21). In both the Icelandic and UKB datasets, polymorphism rate and expected heterozygosity were correlated with the number of bp in the motif (motif length), RRT length, fraction of G/C bases in the STR's motif (motif GC content), and repeat purity[32] (Supplementary note 1). Our microsatellite genotyping is limited by read length (151 bp for most samples), resulting in a decrease in accuracy when detecting and determining genotypes of alleles with long RRTs. While the RRT length limit for inclusion in our study is 140bp, the average expected heterozygosity in our datasets decreases at RRT lengths exceeding 80 bp (Fig. S1,Fig. S2), so we conclude that this is the length where the effect of the RRT length on the genotyping accuracy becomes pronounced.

mDNM rate

We detected 76,987 mDNMs in 6,084 Icelandic trios (mean: 12.7 mDNMs per trio) in 634,406 high quality microsatellites, where the genotypes of the offspring were inconsistent with those of the parents[32]. The trios yielded on average 256,066 microsatellites (40.4%) where all three genotypes passed quality filters and we were able to test for mDNMs.

We estimated the false positive rate of our method between 2.0-5.6% using three approaches; PacBio CCS sequence data, mDNM sharing between monozygotic twins and haplotype sharing across three-generation families (Supplementary note 2).

We estimated the average mDNM rate across all motif lengths as $4.95 \cdot 10^{-5}$ mutations per microsatellite per generation (MMG), consistent with previous results (Supplementary note 5). We observed an order of magnitude difference in mDNM rates between motif lengths, ranging from $1.04 \cdot 10^{-5}$ MMG for hexanucleotide repeats to $1.07 \cdot 10^{-4}$ MMG for dinucleotide repeats (Table 1, Fig. 2)[32]. Using motif length specific mutation rates and the average number of markers available for each trio to extrapolate to the full set of 1,394,292 microsatellites gives an expected number of 63.7 mDNMs (95% CI: 61.9-65.4) per offspring per generation[32]. This extrapolated number is comparable with the de novo mutational load of SNPs and indels detected through short read sequencing[34,35].
We find that mDNMs are depleted within functionally annotated and early replicating regions[32]. We also find that exon intersecting microsatellites contain more interruptions than intergenic ones (Supplementary note 4). Since longer uninterrupted repeats associate with higher mDNM rates[27], this suggests that variants interrupting the repeat structure are positively selected in exons as a way to reduce the occurrence of high impact mDNMs.
We further replicate known mutation rate behaviors such as expansion biases at shorter RRTs and contraction biases at longer RRTs (Supplementary note 3).

Parent of origin effects

To determine the differences between the sexes in mDNM formation, we assigned a parent of origin to 46,171 (60.0%) of the mDNMs using a combination of three methods; read pair tracing, allele sharing and haplotype sharing in three-generation families[32]. The concordance was above 93% between all three methods (Table S11, Table S12, Table S13). We found mDNMs from fathers (N=35,501, 76.9%) to be 3.3 (95% CI: 3.2-3.4, chi squared test $P < 1 \cdot 10^{-320}$) times more common than from mothers (N=10,670, 23.1%) (Table S29).
Maternal and paternal mDNMs occurred with different probabilities at different RRT lengths, motif lengths, and motif equivalence classes. First, a larger fraction of maternal mDNMs

occurred at homopolymers, tri-, penta-, and hexanucleotide microsatellites, whereas a larger fraction of paternal mDNMs occurred at di- and tetranucleotide microsatellites (Table S25, Fig. 3, Supplementary note 7). Second, the average number of bp affected by each mDNM was greater from mothers (3.4 bp, 95% CI: 3.3-3.4) than fathers (3.1 bp, 95% CI: 3.1-3.1) (Mann-Whitney U test $P = 5.0 \cdot 10^{-8}$, Table 3), consistent with previous results[29]. Stratifying by motif length revealed that maternal mDNMs affected more bp than paternal mDNMs on average at homopolymer, di- and tetranucleotide microsatellites (Table S7). Paternal mDNMs occurred at microsatellites with greater RRT lengths. Stratifying by motif length, we observed significant RRT length differences between the sexes at di-, tetra-, penta- and hexanucleotide microsatellites (Table S8).

It seems likely that the sex differences in the accumulation of mDNMs transmitted to offspring are due to differences in the exposure to mutagens of the germlines leading to oocytes and spermatogonia. For di- and trinucleotide microsatellites, the relative frequency of maternal and paternal mDNMs differed between the motif classes (Table S9). An enrichment of CCG motifs in maternal mDNMs may be the result of the vulnerability of GC rich sequences to alkylation[36] or oxidative damage[37] and the long time oocytes spend in dictyate arrest before meiosis. Building on this, we considered all non-trinucleotide mDNMs in microsatellites whose motif contained only G or C and found a 2.3 (95% CI: 1.7-3.7) fold enrichment of maternal mDNMs indicating that purely GC repeats are more prone to mutate in maternal than paternal germlines. Because of a known trend towards paternal expansion at an ATTCT repeat associated with spinocerebellar ataxia 10 (*SCA10*)[38,39], we examined motifs from this class specifically to determine if the bias observed at *SCA10* could be observed genome-wide (Table S9). The maternal contribution at pentanucleotide mDNMs was 25.9% overall, but for the ATTCT motif equivalence class, the ratio was only 18.9% (chi squared test $P = 0.037$). On average, we see an addition of 0.3 bp in paternal mDNMs at microsatellites with ATTCT motifs, compared to a loss of 3.2 bp in maternal mDNMs (Mann-Whitney U test $P = 5.0 \cdot 10^{-4}$). This suggests that the paternal expansion bias observed at the *SCA10* microsatellite also affects other ATTCT microsatellites in the genome.

Parental age influences number of mDNMs

The number of mDNMs transmitted to the offspring was affected by both paternal ($P = 5.4 \cdot 10^{-176}$) and maternal ($P = 7.2 \cdot 10^{-24}$) age at conception[32]. The increase was 0.97 mDNMs per year in fathers (95% CI: 0.90-1.04) and 0.31 mDNMs per year in mothers (95% CI: 0.25-0.37), resulting in an expected number of 51.2 (35.7 paternal and 15.5 maternal) mDNMs and 77.0 (55.1 paternal and 21.8 maternal) mDNMs extrapolated genome-wide in offspring with 20- and 40-year-old parents, respectively. An increase of mDNMs with paternal age is consistent with previous studies[29], but, to our knowledge, this is the first demonstration that mDNMs are also affected by maternal age.

mDNMs at mono-, di-, tri- and tetranucleotide microsatellites increased significantly with paternal age and mDNMs at di-, tetra- and hexanucleotide repeats increase with maternal age (Table 2). An age effect is likely for all motif lengths in both sexes but we lack power to detect it in less frequent motif lengths.

When we consider the fraction of mDNMs of each motif length that each individual carries we see a clear difference in the mutagens acting on the two sexes (Supplementary note 8). The fraction of dinucleotide paternal mDNMs decreased with paternal age, while the fraction of tetranucleotide mDNMs increased with paternal age (Fig. 3). The fraction of hexanucleotide maternal mDNMs increased with maternal age (Fig. 3). The heterogeneity between the sexes in motif length fractions that change with age indicates that, for each sex, the processes behind age related mDNM accumulation are more likely to mutate microsatellites at different motif lengths.

Parental genotypes associate with number of mDNMs in offspring

To determine whether there are variants in the genome that affect the number of mDNMs transmitted by parents to their offspring, we performed a genome-wide association study (GWAS) using the number of mDNMs originating from each parent as a phenotype, adjusting for age, sex, sequencing method and sample type[32].

Three correlated SNPs rs4987188, rs112587140 and rs113983130 (Table S5) were associated with an increase in mDNMs for all motif lengths. We selected [40] rs4987188[A], a missense

variant (p.Gly322Asp, AF=1.9%) in the *MSH2* gene, a mismatch repair gene[41,42], as the lead marker for the association (Fig. 4, Table S6).

Each copy of rs4987188[A] is associated ($P = 3.6 \cdot 10^{-10}$) with a 0.37 s.d. (95% CI: 0.26-0.48) increase in the number of transmitted mDNMs, corresponding to 13.1 and 7.8 mDNMs genome-wide per paternal and maternal copy, respectively. No significant difference in effect was found in the increase between the two parental sexes ($P = 0.14$). However, we see a nominal association ($P = 0.019$) between paternal age and mutation count in offspring of male carriers, suggesting an increase in the effect of rs4987188[A] with paternal age.

The protein encoded by *MSH2* forms two heterodimeric mismatch repair complexes. One predominantly required for repairing mismatched bp and the other mainly responsible for repairing insertion/deletion loops between one and twelve bp[43,44]. Multiple variants in *MSH2* have been reported to cause Lynch syndrome (also known as hereditary non-polyposis colorectal cancer, HNPCC), which results in increased risk of endometrial, colorectal and other cancers[45]. We tested rs4987188[A] for association with an increased risk of endometrial and colorectal cancer in both the Icelandic and combined meta-analysis data sets available at deCODE genetics, but found no such association. Functional studies of the yeast homologue of rs4987188 indicate that it results in a modest decrease in mismatch repair efficiency[46,47].

Homopolymer mDNMs have a higher false positive rate than other mDNM categories. To assess the robustness of our results, we reran the GWAS on mDNM counts per parent without mDNMs occurring at homopolymers. The *MSH2* association was retained with a similar effect of 0.39 s.d. ($P = 1.3 \cdot 10^{-10}$). Furthermore, we find a novel association between mDNM counts without homopolymers and two correlated (r^2=0.988) SNPs, rs8191642 ($P = 5.6 \cdot 10^{-10}$) a synonymous variant (Pro188), and rs8191649 ($1.4 \cdot 10^{-9}$), an intronic variant, both in *NEIL2*, a glycolase involved in both transcription and replication associated base excision repair of DNA damaged by oxidation or by mutagenic agents[48]. We selected [40] rs8191642[G] (AF = 20.3%), as the lead marker for the association.

Testing rs8191642[G] for mothers and fathers separately revealed heterogeneity in effect between the sexes ($P = 1.2 \cdot 10^{-3}$). Each copy of rs8191642[G] is associated with ($P = 7.2 \cdot 10^{-12}$)

a 0.21 s.d. (95% CI: 0.15-0.27) increase in the number of paternally transmitted mDNMs with motif lengths greater than one bp, corresponding to 4.4 mDNMs genome-wide per copy, but associated with a modest increase in the number of maternally transmitted mDNMs (P = 0.015, effect=0.072 s.d.).

To assess the effect of these microsatellite mutator alleles on other types of genetic variation, we compared sDNMs transmitted from carrier and non-carrier mothers and fathers. The results of these comparisons suggest that the effects of rs4987188[A] and rs8191642[G] are confined to microsatellite and they do not otherwise affect the fidelity of the mismatch or base-excision repair pathways (Supplementary note 9).

Discussion

We generated two large microsatellite genotype sets. The microsatellite genotypes for the UKB samples are publicly available and have been tested for association with various phenotypes[49].

In the Icelandic set, we identified 76,987 mDNMs and found an association between the number of mDNMs transmitted to offspring and the age of both mothers and fathers at conception and parental genotypes. We observed a previously unreported increase in the number of maternal mDNMs transmitted to offspring with maternal age, consistent with the increase of both SNP/indel DNMs[34] and recombination with maternal age[24,50]. mDNMs are often associated with replication adducts[26,27,51–54], however the maternal effect gives novel insight into the formation of mDNMs, as oocytes are in dictyate arrest compared to the actively dividing spermatogonia in aging fathers. The maternal age effect indicates that, mDNMs can also occur outside of DNA synthesis during S-phase replication, presumably since DNA polymerases operate during most types of DNA repair on long tracts and in homologous recombination pathways[26].

The observation of different frequencies of mDNM motif classes transmitted by older mothers and fathers allows us to shed light on the mutational processes acting in the germlines. Since spermatogonia undergo a greater number of mitotic cell divisions than oocytes, we propose that the paternal enrichment of AC motif class mDNMs is due to out-of-register realignments

during replication [53], whereas the maternal enrichment of mDNMs at pure GC repeats is likely to be a result of damage accumulated during the dictyate arrest of oocytes.

Sequence variants detected in humans that decrease sequence stability have for the most part been very deleterious and under strong negative selection[55]. Interestingly, the microsatellite mutators at *MSH2* and *NEIL2* identified here are both present in high frequencies and have large effects, thereby indicating that a genome-wide increase in the mDNM rate is not sufficiently deleterious to result in strong selection against these variants. The similar effect of the missense variant in *MSH2* across the sexes, indicates that gametes from both sexes are subject to the same sequence fidelity maintenance process. *NEIL2* has been reported to function in both transcription and replication coupled repair[48] and thus, it is likely that the association between the synonymous variant in *NEIL2* and the number of paternally transmitted mDNMs is due to the more frequent replication of spermatogonia.

We have identified the first germline variants, segregating at high frequencies, that directly affect the mDNM rate in humans. We have also demonstrated for the first time that the number of maternally transmitted mDNMs increases with maternal age. Last, we have generated a publicly available microsatellite genotype set for 150,119 samples, a valuable resource for the scientific community in its efforts to better understand and define the many ways that microsatellites affect human phenotypes.

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Methods

Identification of tandem repeats

To generate a set of STR locations we ran version 4.09 of Tandem repeats finder[31] on the GRCh38[56] human reference genome with the following parameters:
./trf409.linux64 genome.fa 2 7 7 80 10 22 100 –d –h –ngs > trf_out_100

Purity computation

We define the ratio between the observed repeat units in a STR and the number of expected repeat units in a perfectly pure STR as the repeat purity.

$$purity = \frac{observed\ repeats}{E[repeats \mid 100\%\ pure]}$$

Expected heterozygosity computation

We compute expected heterozygosity using the following formula[33]

$$gene\ diversity = \left[\frac{n}{(n-1)}\right]\left(1 - \sum_{i=1}^{k} p_i^2\right)$$

where *n* is the total number of sequences, *k* is the number of alleles at the marker and p_i is the frequency of each allele.

Genotype and marker filtering

For a trio comparison we require all members of the trio to have a phred scaled genotype quality value higher than 60. We removed microsatellites which imputed to a 0% frequency and ones that failed our best practices filters. These filters consider coverage, genotype quality, number of samples genotyped and fraction of reads not supporting the reported genotype and are described in[49].

We further removed microsatellites outside the Tier 1 regions defined by GIAB[57] and microsatellites lying within CNV regions annotated by CNVnator[58].

After using long range phased SNP genotypes to phase and impute our microsatellite genotypes into the Icelandic population[59], microsatellites with alleles showing a strong deviation from the Hardy Weinberg equilibrium ($P < 1 \cdot 10^{-6}$) or a frequency weighted imputation information lower than 0.9 were removed[60]. We also required the frequency weighted phasing information to be greater than 0.6 and the r^2 for leave one out cross validation of phased genotypes to be greater than 0.5.

We looked for offspring with homozygous mDNMs less than 1Mbp apart, implying a haploid state, and checked them for large CNVs covering the region, causing autozygosity and spurious

mDNM calls. Last, we crossed all mDNMs with deletions and duplications imputed into the offspring[61] and removed the ones intersecting.

De novo detection

For marker-trio combinations where all conditions described above are met, we define a mDNM as a trio genotype combination satisfying neither of the following logical statements:

(1) $proband_{A1} \in M \; and \; proband_{A2} \in F$
(2) $proband_{A1} \in F \; and \; proband_{A2} \in M$

where $proband_{A1}$ and $proband_{A2}$ refer to the two alleles carried by the offspring and M and F define the allele pairs carried by the mother and father, respectively.

Microsatellite attribute regression on mDNM rate

We performed a Poisson regression using the number of available markers per trio as an offset on the full data set and stratified by motif length to examine if the effects of other attributes on the mDNM rate remained consistent across motif lengths.

The direction and statistical significance for both RRT length and repeat purity remain consistent for all motif lengths but the effect of GC content in repeat motif is positive for homopolymer repeats (Table S2).

Obtaining confidence intervals for mDNM rate estimates

We used the boot package for R[62,63] to obtain confidence intervals for both our genome wide mDNM rate estimate and the motif length specific estimates. 100 replicates were used in all cases and 95% confidence intervals extracted using the resulting quantiles.

Extrapolation from mDNM rate to expected number of mDNMs

The per motif mDNM rate was determined from our set of high-quality microsatellite calls. We estimated the expected number of per motif mDNMs among all microsatellites by multiplying the mDNM rate and the number of microsatellites per motif length. The total expected mDNMs is the sum over all motif lengths

$$E[mDNMs] = \sum_{i=1}^{6} r_i \cdot n_i$$

where r_i is the mDNM rate at motif length i and n_i is the genome-wide number of microsatellites with motif length i.

mDNM overrepresentation analysis

We counted mDNMs within and outside of regions with various annotations and computed overrepresentation (OR) for each annotation type using the following equation

$$OR = \frac{mDNMs_a / bp_a}{mDNMs_{out} / bp_{out}}$$

Where $mDNMs_a$ enumerates the number of mDNMs within the annotated regions, bp_a represents the total size of the annotated region in bp, $mDNMs_{out}$ enumerates the number of mDNMs outside of the annotated region and bp_{out} represents the number of bp outside the annotated region.

Microsatellite mDNM phasing

We determined the parent of origin of mDNMs, using three distinct methods; read pair tracing, allele sharing and haplotype sharing in three-generation families.

First, we used long range phased[59] SNP and indel genotypes available at deCODE genetics[59] to phase reads reporting mDNMs when possible. Read phases enabled us to assign a parent of origin to mDNMs according to the phase of the reads reporting the event since a read phased to one parent and reporting a mDNM indicates that the mDNM was transmitted from that parent. Not all mDNMs had supporting reads containing phased markers and for these, assigning a parent of origin was not possible (Fig. S5).

Second, for trios where the de novo allele was seen in neither parent and the other offspring allele was only seen in one parent we phased the mDNM to the other parent (Fig. S6).

Last, we used haplotype sharing in three generation families such that if the mDNM was transmitted from offspring to their children, we phased the de novo to the parent sharing a haplotype with its grandchild, and to the parent not sharing a haplotype if de novo was not transmitted (Fig. S7).

Parental age effect regression

We applied a Poisson regression model described in[34] on our data with the number of available markers in each trio as a covariate to determine the effect of parental age at conception on the number of mDNMs transmitted to the offspring.

To determine whether the coefficients were robust to the phasing method, we repeated the analysis[34] and split the phased markers by phasing method and performed the regression on each subset. Both maternal and paternal age remained significant in all subsets (Table S28).

We compute the total predicted number of de novo mutations in a offspring with an X year old father and a Y year old mother from the coefficients from our regression model:

$$(I_F + \beta_F \cdot X + I_M + \beta_M \cdot Y)/(\frac{\mu_{markers}}{1{,}394{,}292})$$

Where I_F is the paternal intercept, β_F is the paternal age effect, I_M is the maternal intercept, β_M is the maternal age effect, $\mu_{markers}$ is the mean value of available markers across our trios and 1,394,292 is the total number of microsatellites we detect.

Construction of mDNM phenotypes

We used maternal and paternal counts at each offspring to construct phenotypes for the parents quantifying the number of mDNMs passed on to the offspring. In addition to correcting for the parental gender (mother/father), we corrected for parental age at birth of offspring, sequencing method, sample type and number of microsatellites available for *de novo* detection in the trio. For parents with more than one offspring in our trio set, we used the average of all their offspring. After regressing out the coefficients, we used rank inverse normalization to normalize the phenotype.

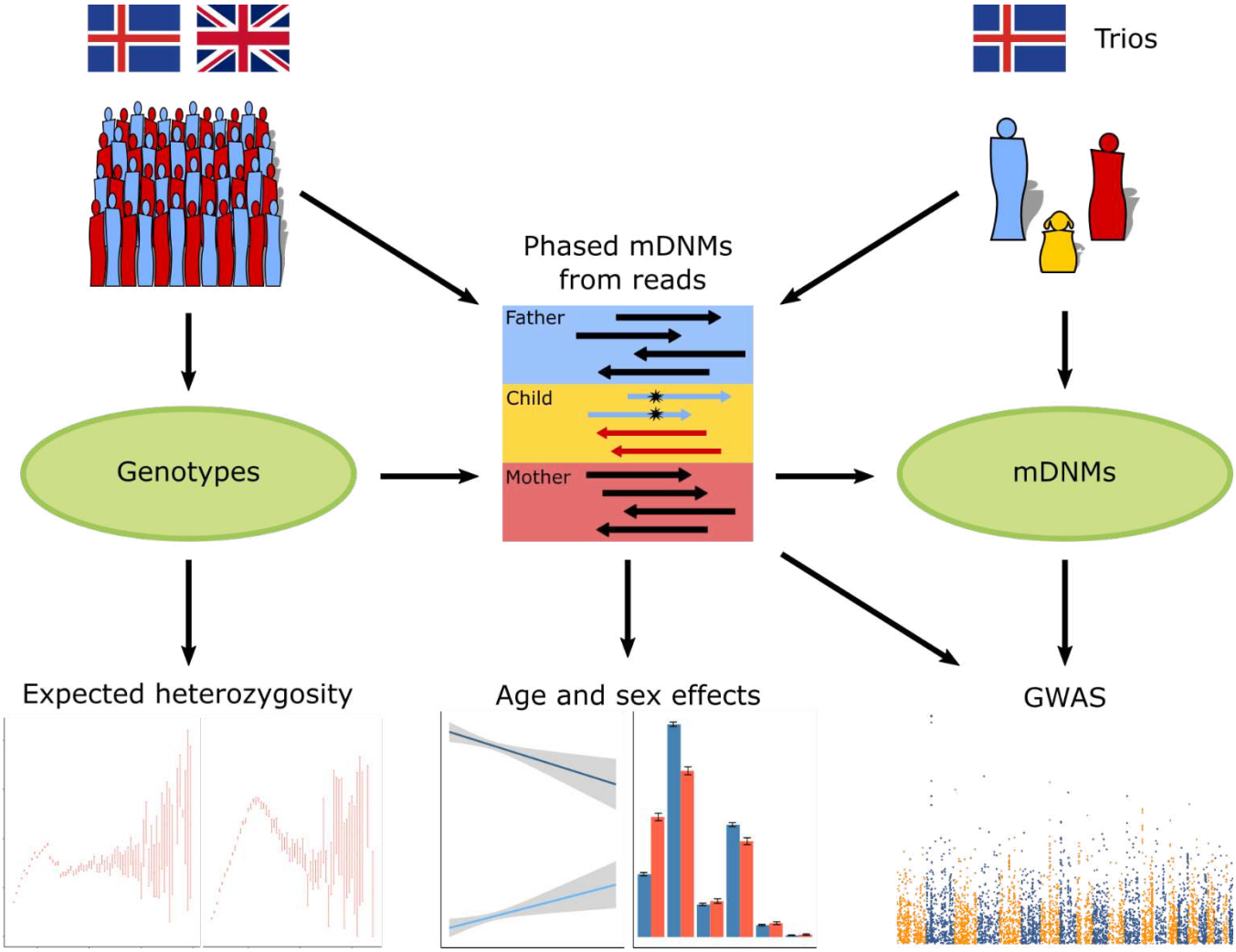

Fig. 1|Overview of the analysis. We use WGS data from Iceland and UKB and genealogy data from Iceland. From the WGS data we generate microsatellite genotypes. Using trios in the genealogy and the genotypes we detect and phase mDNMs and count the number of mDNMs per trio. We associate the individual mDNM counts with genotypes of the parents in the trios. We compute population wide expected heterozygosity based on the genotypes and observe how it is affected by sequence context. From the phased mDNMs we also estimate age and sex effects on the mDNM rate and to create parental phenotypes based on number of mDNMs found in offspring from the phased mDNMs.

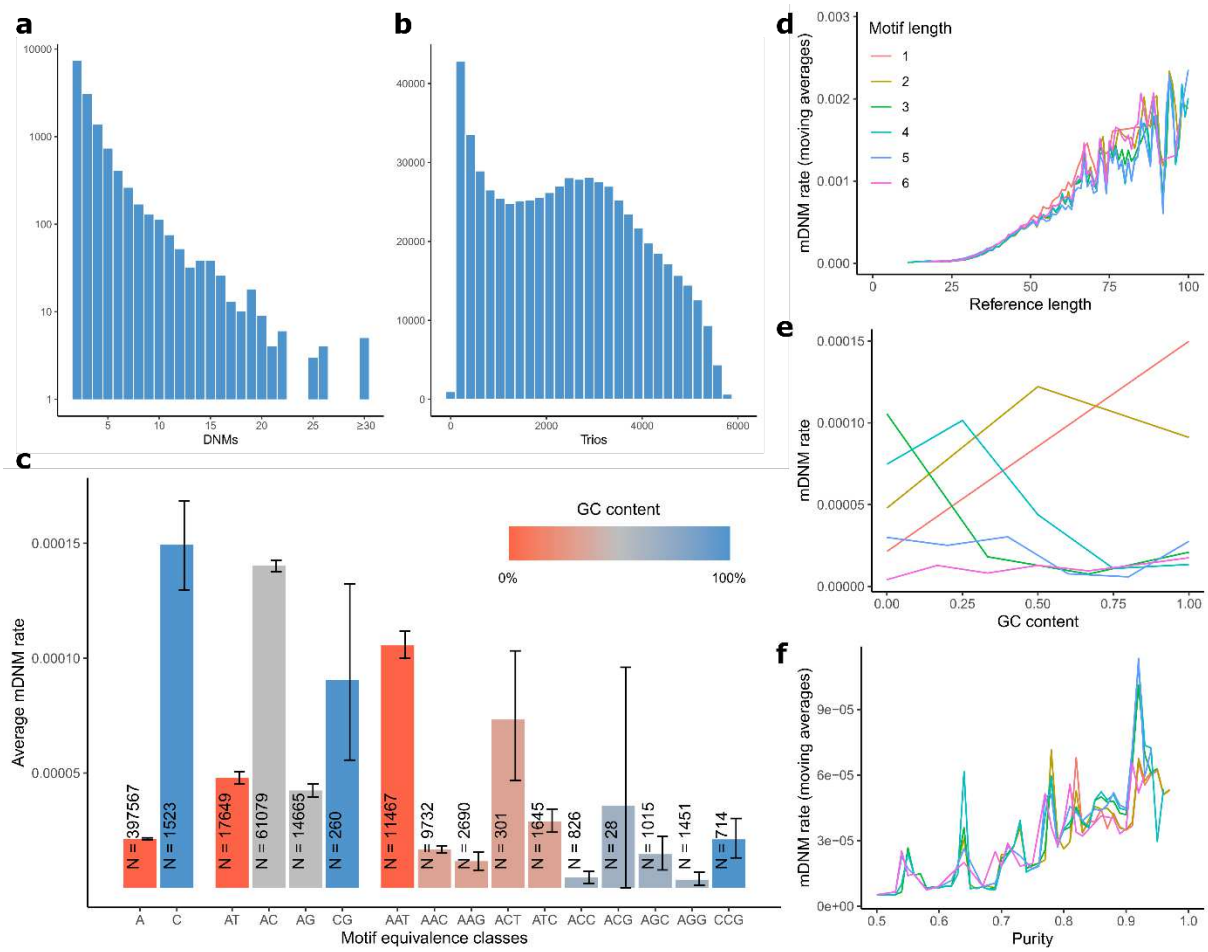

Fig. 2|mDNM rate a) Histogram showing the number of mDNMs per microsatellite. Only microsatellites with at least one mDNM are shown and the counts are on a log-scale. b) Histogram showing the number of trios available to check for mDNMs per microsatellite with at least one mDNM. c) mDNM rates of motif equivalence classes for motif lengths between one and three bp with error bars representing 95% confidence intervals. d) mDNM rate as function of RRT length stratified on motif length. The RRT length affects mDNM rates in a similar way for all motif lengths. e) mDNM rate as function of GC content in motif. f) mDNM rate as a function of repeat purity stratified on motif length. The mDNM rate increases with purity for all motif lengths.

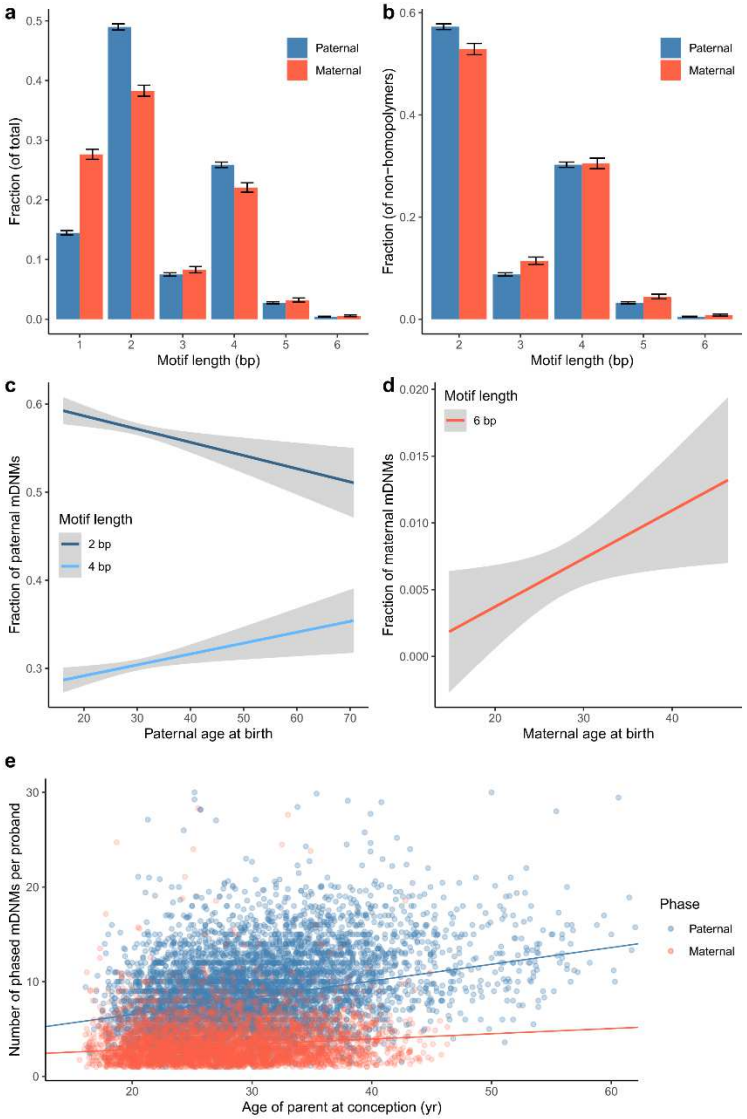

Fig. 3|Sex differences and age effects a) Relative frequencies of paternal and maternal mDNMs at different motif lengths. b) Relative frequencies of paternal and maternal mDNMs at different motif lengths without homopolymer mDNMs. c) Fraction of di- and tetranucleotide paternal mDNMs as a function of paternal age. The tetranucleotide fraction increases while the dinucleotide fraction decreases. d) The fraction of hexanucleotide maternal mDNMs increases with maternal age. e) Paternal and maternal age effect regression lines within our mDNM set. The age effect reported is interpolated to a genome-wide value using the fraction between average number of available microsatellites and total number of microsatellites. Fractions in c) and d) are computed after excluding homopolymers.

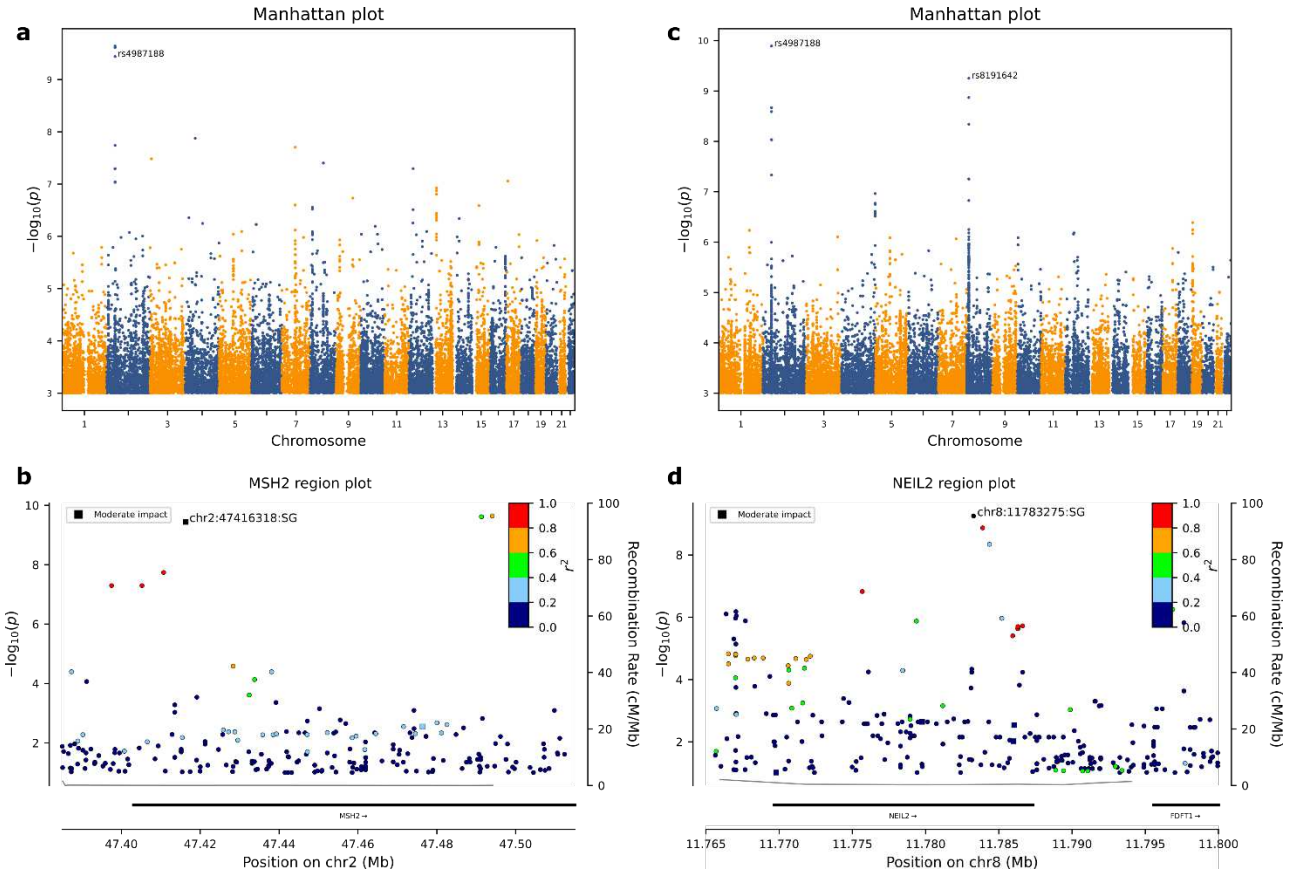

Fig. 4|Genome-wide association a) Manhattan plot showing a missense variant in the *MSH2* gene associating with an increased number of mDNMs transmitted to offspring. b) Region plot for the missense variant in *MSH2*. Two correlated markers also reach genome-wide significance. c) Manhattan plot showing a synonymous variant in the *NEIL2* gene which associates with an increased number of mDNMs with motif lengths greater than one bp transmitted to offspring. The plot also shows the p-value for the *MSH2* missense variant considering mDNMs with motif lengths greater than one bp. d) Region plot for a synonymous variant in the *NEIL2* gene. No other markers at the locus reach genome wide significance.
Chromosomes on Manhattan plots are marked with alternating colors and the threshold for genome wide significance was set as $1 \cdot 10^{-9}$.

Motif (bp)	mDNM rate (95% CI)	#microsatellites	#DNMs
1	$2.15 \cdot 10^{-5}$ ($2.11 \cdot 10^{-5}$-$2.19 \cdot 10^{-5}$)	399,087 (62.9%)	17,194 (22.3%)
2	$1.07 \cdot 10^{-4}$ ($1.06 \cdot 10^{-4}$-$1.09 \cdot 10^{-4}$)	93,653(14.8%)	33,025 (42.9%)
3	$4.92 \cdot 10^{-5}$ ($4.70 \cdot 10^{-4}$-$5.21 \cdot 10^{-4}$)	29,869 (4.7%)	5,469 (7.1%)
4	$8.57 \cdot 10^{-5}$ ($8.28 \cdot 10^{-4}$-$8.86 \cdot 10^{-4}$)	66,541 (10.5%)	18,795 (24.4%)
5	$2.58 \cdot 10^{-5}$ ($2.37 \cdot 10^{-5}$-$2.86 \cdot 10^{-5}$)	29,992 (4.7%)	2,132 (2.8%)
6	$1.04 \cdot 10^{-5}$ ($9.07 \cdot 10^{-6}$-$1.23 \cdot 10^{-5}$)	15,264 (2.4%)	372 (0.5%)
Total	$4.95 \cdot 10^{-5}$ ($4.88 \cdot 10^{-5}$-$5.02 \cdot 10^{-5}$)	634,406	76,987

Table 1: mDNM rate, number of microsatellites in high quality set and number of mDNMs, all by motif lengths. Dinucleotide repeats have the highest mDNM rate and represent almost 43% of our mDNMs.

Data set	Paternal age effect (95% CI)	Maternal age effect (95% CI)	#paternal/#maternal
All markers	**0.178(0.165-0.190)**	**0.058(0.047-0.069)**	**35,501/10,670**
Motif length 1	**0.013(0.006-0.020)**	0.007(-0.001-0.014)	5,141/2,948
Motif length 2	**0.078(0.069-0.087)**	**0.029(0.022-0.037)**	**17,388/4,084**
Motif length 3	**0.011(0.008-0.014)**	0.002(-0.001-0.004)	2,663/883
Motif length 4	**0.056(0.050-0.062)**	**0.013(0.008-0.017)**	**9,184/2,354**
Motif length 5	0.004(-0.003-0.011)	-0.002(-0.007-0.003)	977/341
Motif length 6	-0.008(-0.021-0.005)	**0.016(0.003-0.029)**	**148/60**

Table 2: Maternal and paternal age effect for all motif lengths and conditioning on motif length. mDNMs at mono-, di-, tri- and tetranucleotide repeats increase with paternal age while mDNMs at di- and tetra- and hexanucleotide repeats increase with maternal age. Bold represents significant association ($P < 0.05$).

Mutation size (In motifs)						
Motif	<-2 (p/m)	-2 (p/m)	-1 (p/m)	1 (p/m)	2 (p/m)	>2 (p/m)
1	312(168/144)	140(69/71)	2,782(1,671/1,111)	3,962(2,777/1,185)	214(121/93)	679(335/344)
2	610(399/211)	2,482(2,101/381)	7,124(6,192/932)	8,933(6,936/1,997)	1,849(1,490/359)	474(270/204)
3	90(52/38)	160(117/43)	1,516(1,053/463)	1,524(1,251/273)	228(171/57)	28(19/9)
4	134(90/44)	237(170/67)	5,229(3,972/1,257)	5,798(4,861/937)	107(73/34)	33(18/15)
5	14(11/3)	44(30/14)	583(392/191)	651(533/118)	13(7/6)	13(4/9)
6	0(0/0)	4(3/1)	79(50/29)	117(90/27)	6(3/3)	2(2/0)
Total	1,160(720/440)	3,067(2,490/577)	17,313(13,330/3,983)	20,985(16,448/4,573)	2,417(1,865/552)	1,229(648/576)

Table 3: Counts of mutation sizes of phased mDNMs per motif length and for all motif lengths combined. Maternal and paternal counts are given in brackets. Maternal mDNMs are larger on average than paternal ones and we see the ratio of maternal mDNMs increases with their size.

Supplementary information

Supplementary note 1: STR properties correlate with polymorphism rate and expected heterozygosity

A number of STR properties correlate with the polymorphism rate and expected heterozygosity[33]. Both the motif length and how many times the motif is repeated in its RRT (repeat number) affect the polymorphism rate and the expected heterozygosity. Almost all homopolymers are polymorphic and both the fraction of polymorphic STRs and expected heterozygosity decrease with motif length (linear regression $P < 1 \cdot 10^{-320}$ for both datasets, Fig. S1, Fig. S2). For all motif lengths, an increase in repeat number results in both a higher polymorphism rate and expected heterozygosity (linear regression $P < 1 \cdot 10^{-320}$ for both datasets, Fig. S1, Fig. S2).

The fraction of G/C bases in an STR's motif (motif GC content) is negatively correlated with polymorphism rate (Table S21) while negative correlation with expected heterozygosity is only observed for motif lengths above two bp (Table S20). Homopolymers from the C motif class have higher expected heterozygosity than A class homopolymers (Table S20) but account for only 0.8% of all homopolymers (Table S15). CpG microsatellites (CG motif class) also have on average higher expected heterozygosity than the other dinucleotide motif classes but account for only 0.4% of all dinucleotide repeats (Table S15, Table S20). Thus, although C class homopolymers and CG class dinucleotide microsatellites have higher expected heterozygosity values than other classes with equal motif lengths, their overall effect on microsatellite diversity is small since they are so rare.

Enrichment of A motif class homopolymers in the human genome is thought to be a result of the microsatellite-like structure often found at 3' ends of reverse transcribed RNA sequences, i.e. poly-A tails[64,65] and the high expected heterozygosity rate at CG class microsatellites is consistent with CpG sites acting as mutational hot spots[66]. Negative correlation of GC content to polymorphism rates and expected heterozygosity can likely be explained by the three hydrogen bonds between paired G/C bp, compared to two between A/T bp, making the slippage-causing disassociation from the template strand during replication less likely.

We define repeat purity as the ratio between the number of times the STR's repeat motif is observed in its RRT and the maximum number of repeat motifs if the sequence contained no interruptions (where the highest possible repeat purity is 1). Stratified on RRT length, all correlations between repeat purity and both expected heterozygosity and polymorphism rate were positive (Table S18, Table S19, Fig. S3, Fig. S4). This is consistent with[67–70], reporting a positive correlation between repeat purity and mDNM rate. Interruptions of the repeat sequence decrease the number of locations where replication slippage can occur and thus the mDNM rate[27].

Supplementary note 2: False positive estimation

To estimate the false positive rate of our method we used two methods; PacBio CCS sequence data, available for four of our trios, and mDNM sharing in nine Icelandic monozygotic twin pairs in our set. We used haplotype resolved assemblies of the PacBio data and while we were unable to verify homopolymer mDNMs as the PacBio sequencing error rate was too high[71] we were able to verify the existence of 27 mDNMs with motif length >1. Out of these, 26 were true positives and one was a false positive at a dinucleotide repeat, giving an expected false positive rate of 3.7% for motif lengths greater than 1. (Table S4).

For mDNMs observed in offspring with a monozygotic twin also present in our set, we checked whether the mDNM genotypes were concordant between both twins. We compared mDNM calls where the genotype was present in the monozygotic twin of the offspring. We treated genotype calls of the other twin present if the genotype quality was higher than or equal to 30, which is half of the value we require for trio mDNM detection. Out of the 230 comparable MZ-twin mDNMs, 217 were found in both twins and 13 were discordant (Table S30) which gives a false positive rate estimate of 5.6%. We note that this is likely to be an overestimate, as some of the differences between the twin pairs could be due to result of post zygotic mutations, representing true differences between twins[72].

Using haplotype sharing across 540 three-generation families (795 trios), we counted how many times an mDNM was transmitted from an offspring to its child and estimated the transmission rate. The expected value of the transmission rate is 0.50 and deviations from it

quantify false positive mDNM detection rates. For example, if the observed mutations were somatic and thus false positive as mDNMs, we would not observe transmission from the offspring to its child. We observe a transmission rate of 0.49 (N = 11,228, 95% CI: 0.48-0.50) which gives an estimated false positive rate of 2%, although transmission rates vary between motif lengths and thus the error rate estimates as well (Table S14). Notably, the transmission rate for homopolymers is only 0.4 while the other motif lengths have transmission rates much closer or equal to 0.5.

Supplementary note 3: Replication of mutation rate trends

As the RRT length increased from ten to 100 bp the mDNM rate also increased for all motif lengths (Fig. 2), consistent with findings from previous studies[26,73–75] and, intuitively, the opportunities for errors during replication increase with a microsatellite's RRT length.

Microsatellites with longer RRTs have been shown to be more likely to contract and shorter ones to expand[26,76–81]. We replicate this. For all RRT length thresholds, a higher fraction of mDNMs had a gain of repeat motifs below the threshold than above it, i.e., the microsatellite group with shorter overall RRTs were more likely to expand than the group with longer RRTs (Table S26).

The mDNM rate for homopolymers was positively correlated with the motif G/C content, while di-, tri-, tetra- and pentanucleotide repeats had a negative correlation with the mDNM rate and for hexanucleotide repeats we lacked power to detect a correlation with the mDNM rate (Fig. 2, Table S2).

Repeat purity correlated positively with the mDNM rate for all motif lengths (effect = 0.12, $P < 1 \cdot 10^{-320}$, Fig. 2, Table S2), consistent with previously published results[75]. The correlation remained positive in most cases after conditioning on RRT lengths (Table S3).

mDNM rates varied by repeat motif class, AAT motif mDNMs were by far the most common ones among trinucleotides. The rare C homopolymers mutated more frequently than the A homopolymers and AC motif microsatellites had the highest mutation rate of the dinucleotide

microsatellites before correcting for RRT length and CG motif microsatellites after correction (Supplement, Fig. 2, Table S1).

Supplementary note 4: mDNMs in functionally annotated and early replicating regions

Previous studies have reported increased efficiency of mismatch repair (MMR) in early-replicating regions of the human genome[82]. Our results are in line with this since we see 1.28 (95% CI: 1.25-1.31) fold depletion of mDNMs in early replicating regions of the genome[83].

Exonic mDNMs are rarer than their intergenic and intronic counterparts. In 2,568,858 transmissions of microsatellites intersecting exons by one or more bp, we observed 33 mDNMs.

We estimated the exonic mDNM rate as $1.3 \cdot 10^{-5}$ MMG, which is 3.9 (95% CI: 2.8-5.6) times lower than the genome-wide estimate. mDNMs are further 1.7 (95% CI: 1.3-2.1), 1.4 (95% CI: 1.3-1.5) and 4.2 (95% CI: 1.2-34.4)-fold depleted in 5'UTR and 3'UTR and splice regions, respectively. The 33 exonic mDNMs occurred at 21 unique microsatellites, of which 19 had motif lengths that were multiples of three and since amino acids are coded with three bp codons, mutations at microsatellites with multiple of three motif lengths are unlikely to cause a frameshift but rather an in-frame alteration of a gene. Sixteen of the exonic mDNMs were trinucleotide microsatellites, three were hexanucleotide microsatellites and the remaining two were homopolymers.

Tri- and hexanucleotide repeats were enriched in coding exons (chi squared test $P < 1 \cdot 10^{-320}$) compared with the rest of genome. Microsatellites with motif lengths that are multiples of three accounted for 93.3% of exon intersecting microsatellites and 70.9% of all exon intersecting non polymorphic STRs (Table S16). In contrast, 12.3% of all microsatellites had motifs that are multiples of three (Table S16) and 44.5% of non-polymorphic STRs.

The average purity of microsatellites was 0.94, while among microsatellites in exons the purity was notably lower (0.87, Mann-Whitney $P = 1 \cdot 10^{-153}$). Purity was positively correlated with the mDNM rate, so decreased purity in exons may decrease occurrences of possibly pathogenic mDNMs. This indicates that there is a possible positive selection for point mutations that

reduce the purity of exonic microsatellites or a possible negative selection for point mutations that increase in their purity. The purity difference is largest for trinucleotide repeats, the most common motif length for exon intersecting microsatellites (Table S17). Non-polymorphic coding STRs do not have decreased purity compared to their intergenic counterparts, so point mutations are less likely to be the mechanism preventing mutations at these exonic STRs (Table S17).

Supplementary note 5: mDNM rate estimate comparison

Our mDNM rate estimate is nominally lower than a previous estimate[29] of $5.6 \cdot 10^{-5}$ and lower than the two estimates of $10.0 \cdot 10^{-4}$ and $2.7 \cdot 10^{-4}$ for tetra- and dinucleotide repeats, respectively[1]. This apparent discrepancy could be a result of a more conservative filtering in our study, a younger set of parents, a generally healthier cohort and a different range of motif lengths considered. However, the most likely reason for the apparent discrepancy is the sample size difference between the studies. Our set contains 53,026 individuals while the set analyzed by Mitra et al. [29] contained 6,548 individuals. Thus, our minimum detection frequency is $1/(2 \cdot 53{,}026) = 9.0 \cdot 10^{-6}$ compared to the minimum detection frequency[29] of $1/(2 \cdot 6{,}548) = 7.6 \cdot 10^{-5}$ enforced by the smaller sample size. We recomputed our mutation rate estimate conditioning on microsatellite frequency (Table S27) and confirmed that at a detection frequency cutoff of $7.6 \cdot 10^{-5}$ our estimate becomes $5.6 \cdot 10^{-5}$ and matches the one presented by Mitra et al.[29]. Similarly, at a minimum frequency of 10% our estimate is comparable to the one from Sun et al.[1] Based on this we conclude that a mDNM rate estimate depends on the size of the sample set studied.

Supplementary note 6: mDNM rate comparison between motif equivalence classes

The mDNM rate is higher for C class homopolymers than for A class ones (Mann-Whitney U test $P < 1 \cdot 10^{-230}$, Fig. 2), but C homopolymers are much rarer and represent only 0.8% of all homopolymers in our set (Table S15). The AC motif class has the highest mDNM rate of the dinucleotide microsatellites (Table S22, Fig. 2). However, the average RRT length of the AC

motif class is longest among dinucleotide classes. Including RRT length as a covariate the CG motif class has a higher mDNM rate than all other dinucleotide classes (Table S23), in line with the fact that CpG sites have been shown to act as mutational hot spots[66].

The AAT motif class had a higher mDNM rate than eight of the other nine trinucleotide repeat motif classes. Only the rarest class (ACG) did not show a significantly different mDNM rate (Table S24). The AAT motif class accounts for 38.4% of all trinucleotide microsatellites and 79.0% of trinucleotide mDNMs and has an mDNM rate 1.5 times higher than the second highest class (Table S23). A higher mDNM rate for AAT class motifs has been previously reported for other organisms[84,85] but not, to our knowledge for, humans.

An in vitro study of how repeat motifs affect the frequency of polymerase slippage during replication reported that motifs less likely to stall replication were more likely to mutate during replication. Of the dinucleotide repeat classes, microsatellites from the AC class had the lowest replication stall affinity[53]. The higher mDNM rate of microsatellites in the AAT motif class could be a result of its low replication stall affinity[53]. The two hydrogen bonds between A/T base-pairs compared to the three between G/C base-pairs also makes A/T pairs more likely to disassociate from each other, enabling the formation of secondary structures and possible mDNMs. Finally, repeats with a high A/T-content also have a sequence composition similar to elements involved in DNA unwinding at replication origins[86] during mitosis. These repeats could therefore function as aberrant replication origins and cause a higher mDNM rate during replication in S phase[86].

Supplementary note 7: Motif length enrichment excluding homopolymers

A higher fraction of maternal mDNMs occur at tri-, penta- and hexanucleotide microsatellites while dinucleotide microsatellites represent a larger fraction of paternal mDNMs (Table S25, Fig. 3).

The average number of bp involved without homopolymers is larger in maternal mDNMs than in paternal mDNMs (3.9 vs 3.4 bp, Mann-Whitney U test $P = 2.6 \cdot 10^{-23}$). Stratifying on motif length reveals that maternal mDNMs affect more bp on average at di- and tetranucleotide microsatellites (Table S7).

Considering mDNMs with motif lengths above one, the number of repeats in the reference is higher for paternal mDNMs (16.8 vs 15.7 repeats, Mann-Whitney U test $P = 2.8 \cdot 10^{-51}$).

Supplementary note 8: Motif length fraction change with age including homopolymers

Tetranucleotide mDNMs increase their fraction with paternal age (Linear regression $P = 1.5 \cdot 10^{-6}$) and the fraction of di- and hexanucleotide mDNMs increases with maternal age.

For both maternal and paternal mDNMs the fraction of mDNMs at homopolymers decreases with age.

Supplementary note 9: Effects on microsatellite mutator alleles on sDNMs

Somatic and germline mutations in mismatch-repair genes are known to cause microsatellite instability and hypermutator phenotypes in tumors of the colon and endometrium. Sequencing of large cohorts of tumors has revealed several mutational signatures associated with mismatch repair deficiency[87].

These signatures show a strong correlation with somatic microsatellite instability. Mutations in components of the base-excision repair pathway, most notably *NTHL1* and *MUTYH,* also cause distinct mutational signatures in tumors with these mutations (COSMIC signatures SBS30 and SBS36[87], respectively). Given the effects of rs4987188[A] and rs8191642[G] on the rate of mDNMs, we were interested in knowing if they also affect the mutational spectra of single-base-substitution DNMs, for example by altering the MMR/BER pathways in the testis.

Using only phased DNMs, we compared the 96-class trinucleotide spectra of DNMs transmitted from carrier and non-carrier mothers and fathers. All spectra were highly similar (pairwise cosine similarities >0.95) and showed no hint of MMR deficient related mutational signatures (Fig. S8). This suggests that the effects of rs4987188[A] and rs8191642[G] are confined to microsatellite sites and they do not otherwise affect the fidelity of the mismatch or base-excision repair pathways.

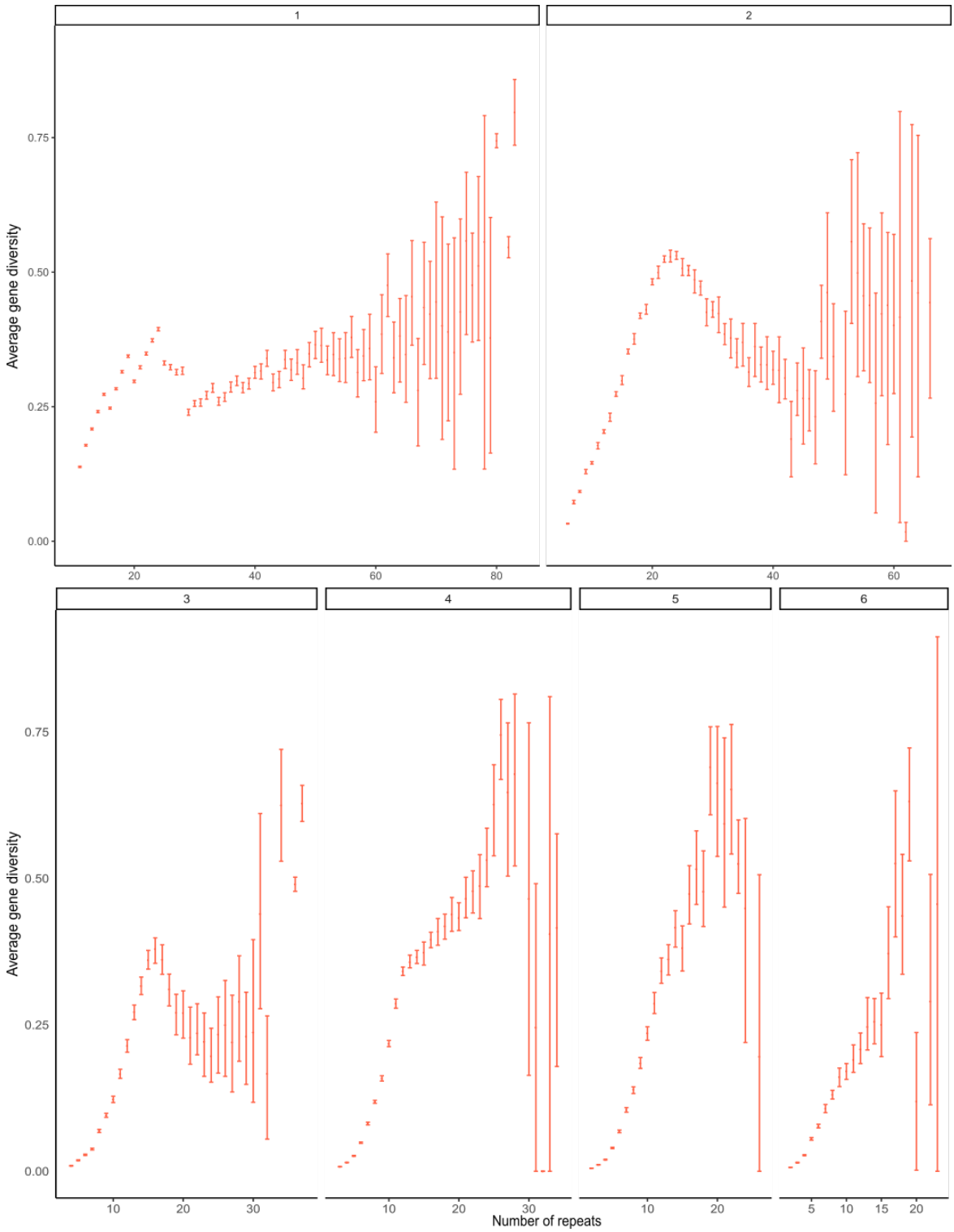

Fig. S1 Average expected heterozygosity in UKB as a function of repeat number stratified on motif length. The drop in all motif lengths is most likely due our inability to reliably detect long alleles from short reads, causing underestimation of expected heterozygosity values at microsatellite with long reference alleles.

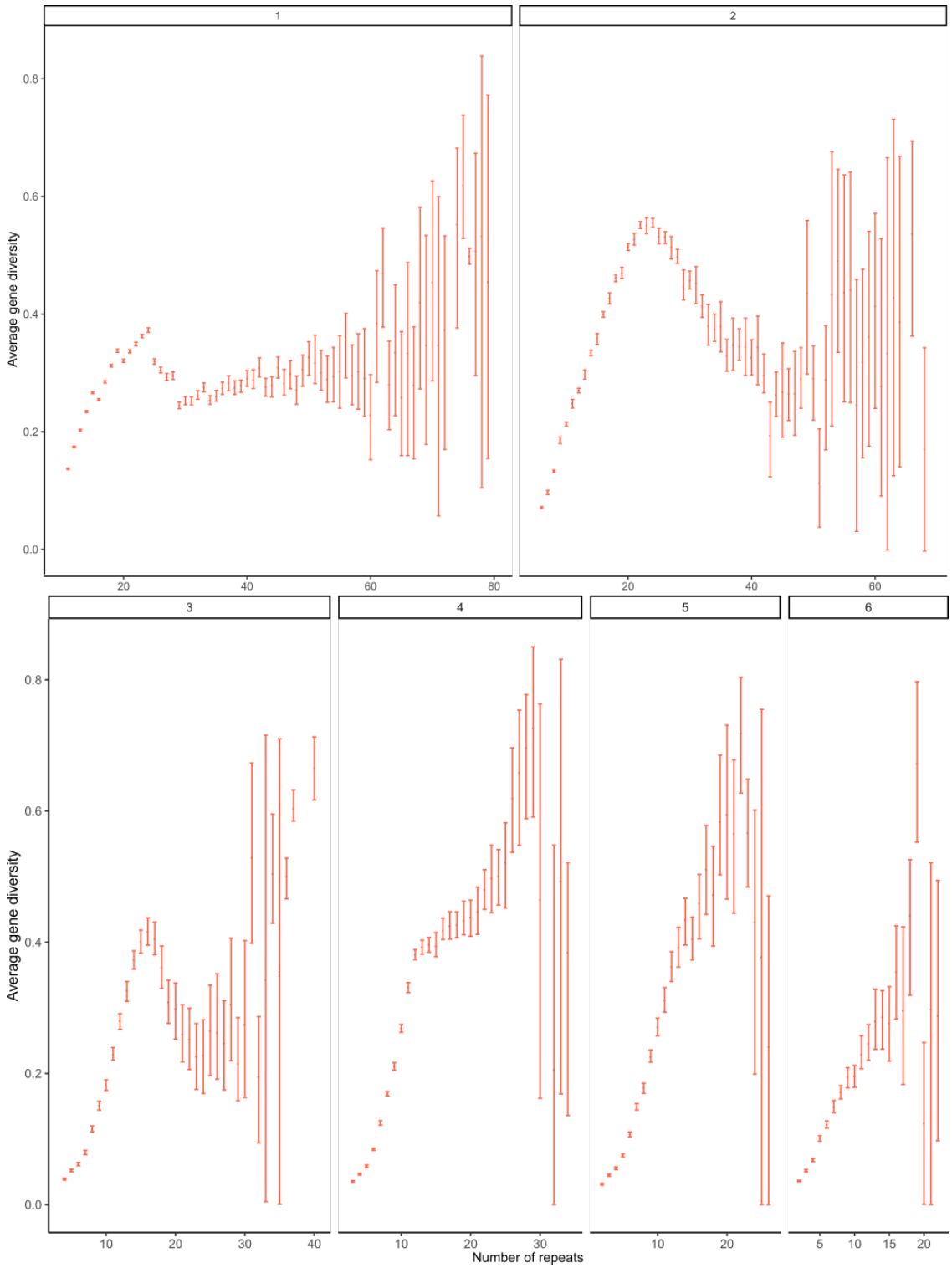

Fig. S2 Average expected heterozygosity in Icelandic data set as a function of repeat number stratified on motif length.

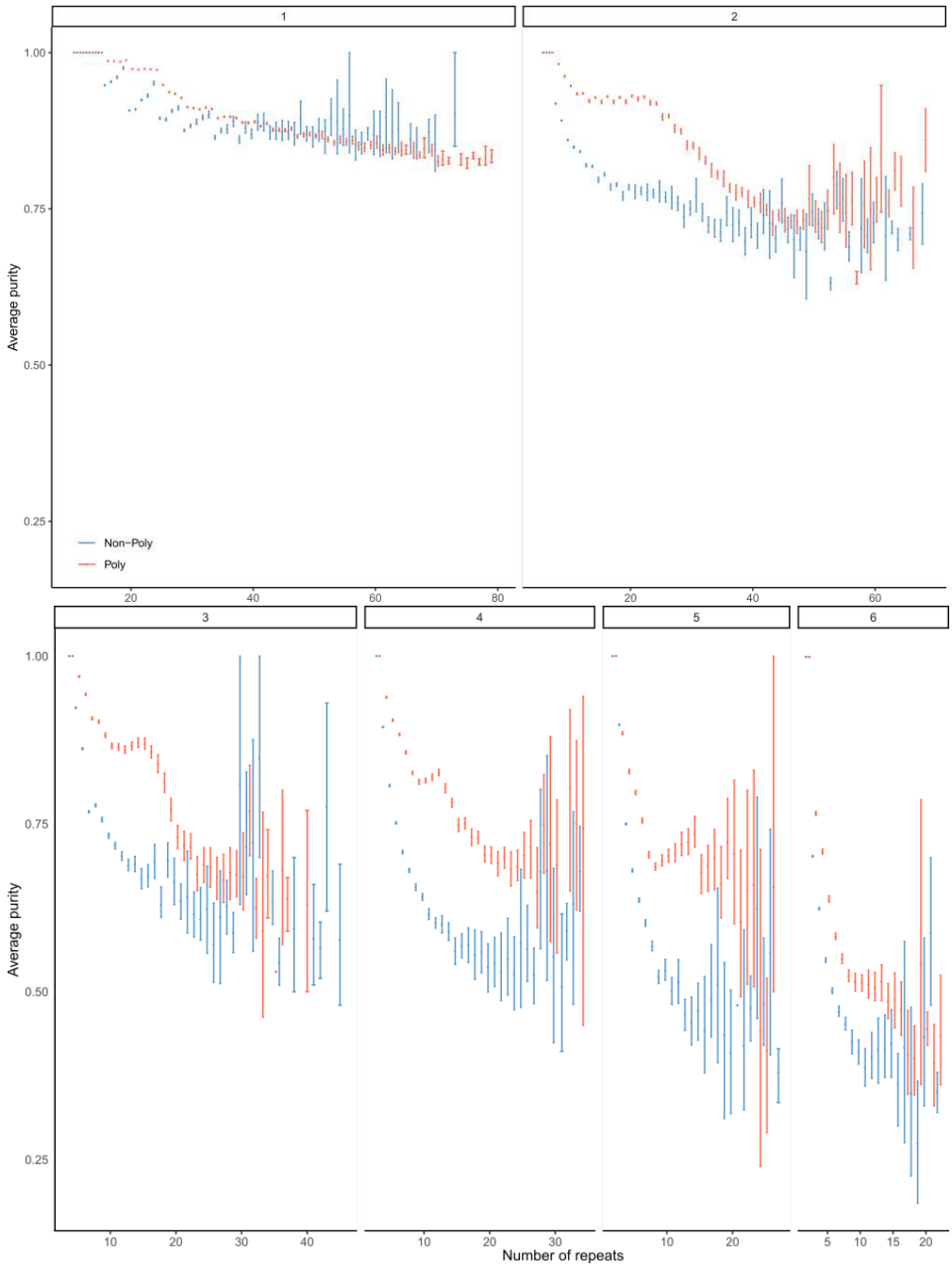

Fig. S3 Average repeat purity as a function of repeat number stratified on motif length for the Icelandic data set. Red marks polymorphic STRs (microsatellites) and blue non-polymorphic STRs.

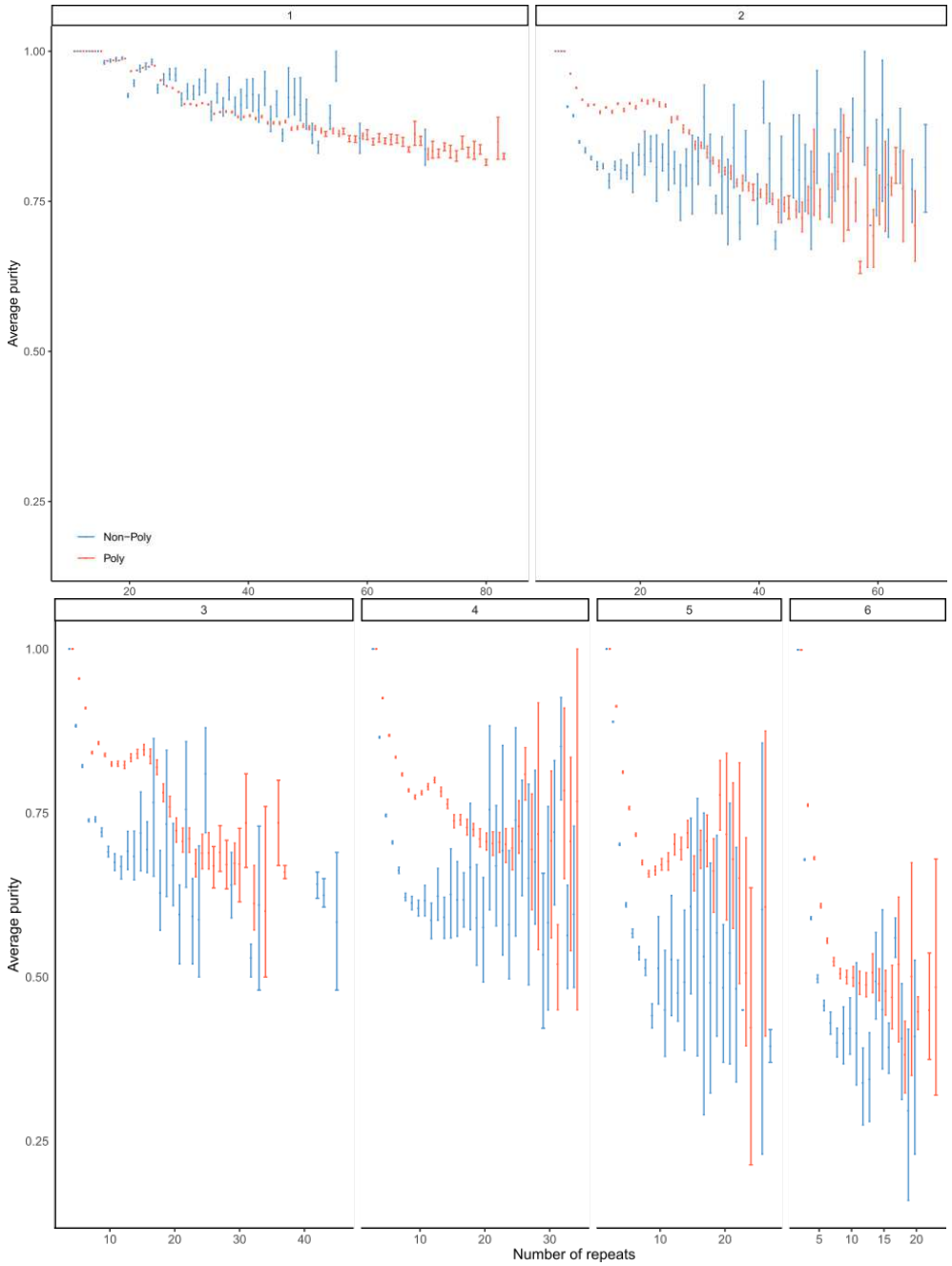

Fig. S4 Average repeat purity as a function of repeat number stratified on motif length for the UKB data set. Red marks polymorphic STRs (microsatellites) and blue non-polymorphic STRs.

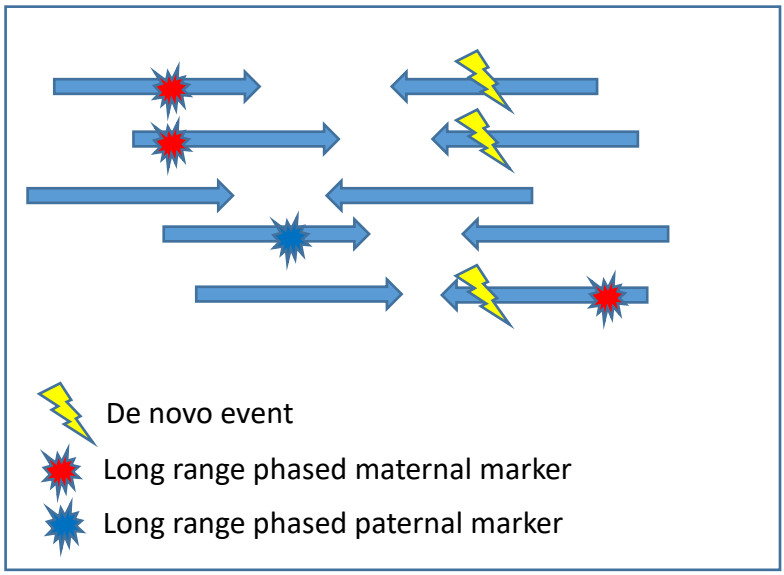

Fig. S5 Read pair phasing. We use read pairs which contain a long range phased marker and report an mDNM. Reads with long range phased markers covering the de novo site but supporting the other allele can also further give information on the parent of origin. An example of a maternal mDNM where three read pairs report a long range phased maternal marker and a de novo allele, one pair is not informative and one contains a long range phased paternal marker and not the de novo allele.

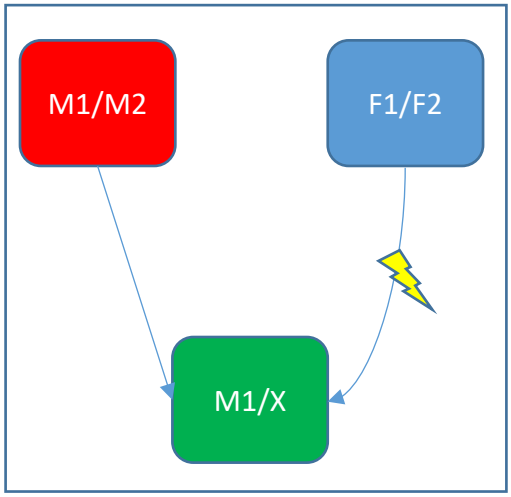

Fig. S6 Allele based phasing: If the de novo allele is present in neither parent and the other allele is present only in one parent we phase the de novo event to the other parent. Here we phase the de novo event to the father since a maternal allele is seen in the offspring but neither of the paternal ones.

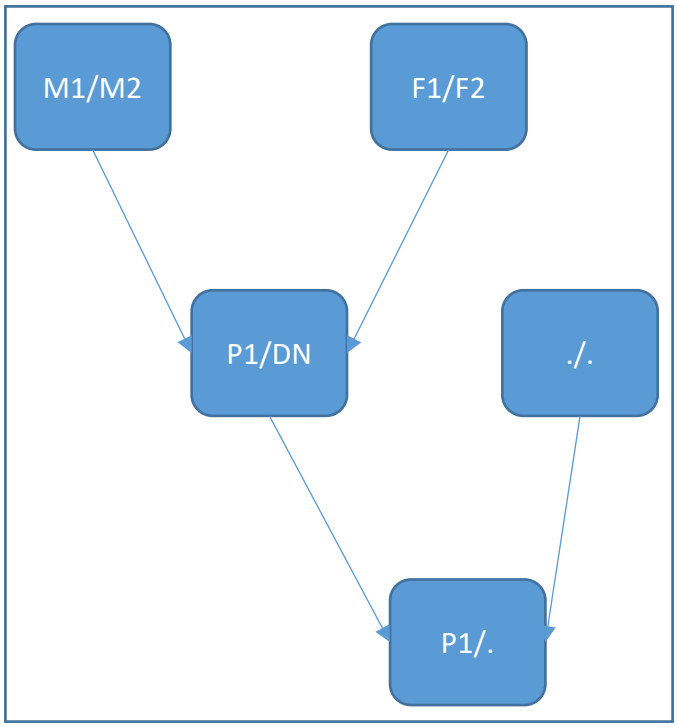

Fig. S7 Haplotype sharing in three generation families: If the de novo is transmitted from the offspring to its child we phase the de novo to the parent sharing a haplotype with the offspring, and to the parent not sharing a haplotype if de novo is not transmitted.

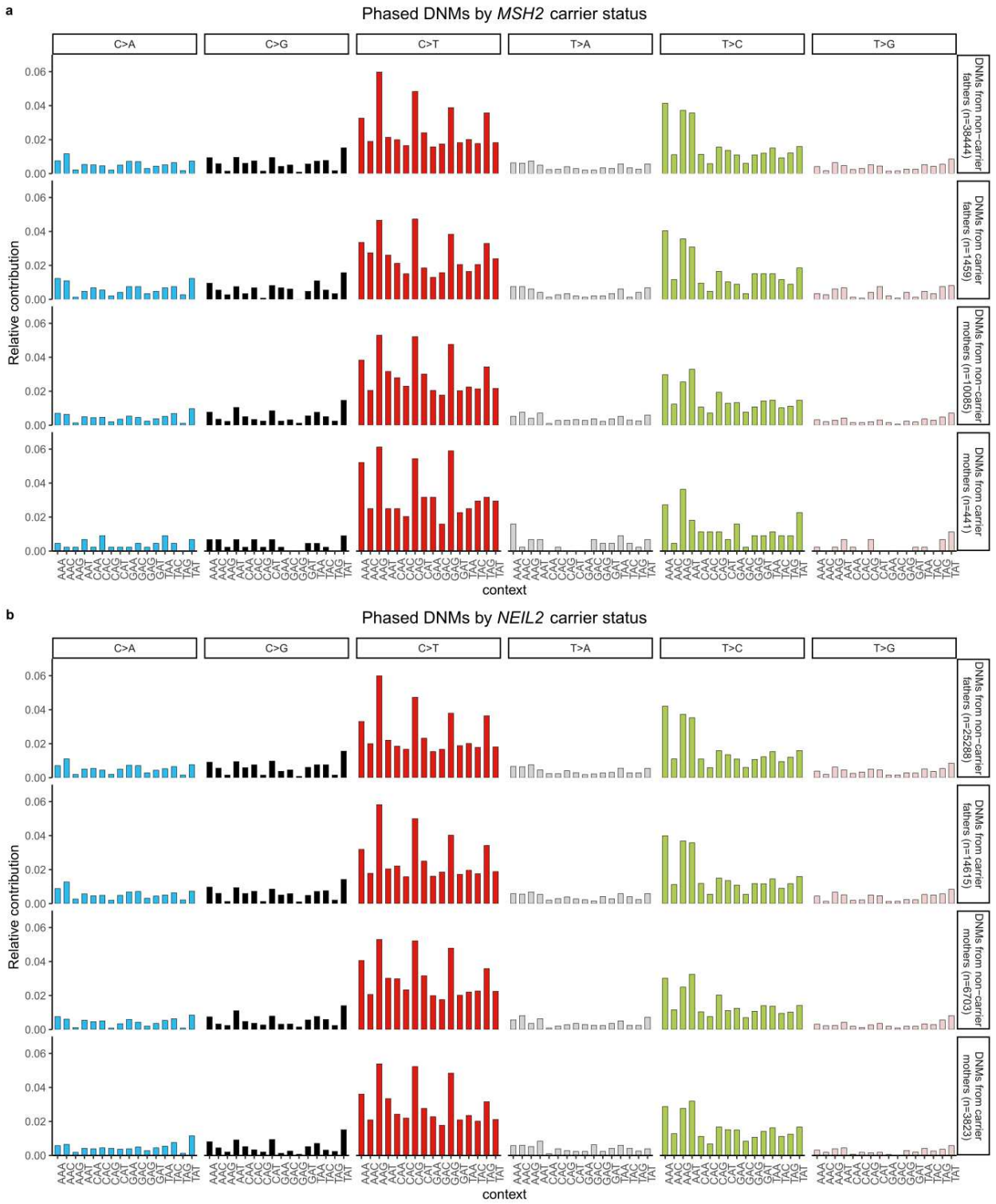

Fig. S8: sDNM mutation signatures for offspring of paternal and maternal carriers and non-carriers of both rs4987188 and rs8191642. The signatures are not different in any of the four comparisons so we assume that the variants do not affect the generation of sDNMs.

Class representative	Other members
A	T
C	G
AC	CA, GT, TG
AG	GA, CT, TC
AT	TA
CG	GC
AAC	ACA, CAA, GTT, TGT, TTG
AAG	AGA, GAA, CTT, TCT, TTC
AAT	ATA, TAA, ATT, TAT, TTA
ACC	CAC, CCA, GGT, GTG, TGG
ACG	CGA, GAC, CGT, TCG, GTC
ACT	CTA, TAC, AGT, TAG, GTA
AGC	GCA, CAG, GCT, TGC, CTG
AGG	GAG, GGA, CCT, CTC, TCC
ATC	TCA, CAT, GAT, TGA, ATG
CCG	CGC, GCC, CGG, GCG, GGC

Table S1 Motif equivalence classes and their members.

Motif	Repeat tract length	Purity	GC content	Motif length
All	0.10 ($<1 \cdot 10^{-320}$)	0.12($<1 \cdot 10^{-320}$)	$9.47 \cdot 10^{-5}$(0.58)	-0.15 ($<1 \cdot 10^{-320}$)
1bp	0.16($<1 \cdot 10^{-320}$)	0.28($<1 \cdot 10^{-320}$)	0.02($<1 \cdot 10^{-320}$)	X
2 bp	0.10($<1 \cdot 10^{-320}$)	0.17($<1 \cdot 10^{-320}$)	-0.01($8.6 \cdot 10^{-87}$)	X
3 bp	0.12 ($<1 \cdot 10^{-320}$)	0.14 ($<1 \cdot 10^{-320}$)	-0.03($3.0 \cdot 10^{-235}$)	X
4 bp	0.10($<1 \cdot 10^{-320}$)	0.10 ($<1 \cdot 10^{-320}$)	-0.003($2.2 \cdot 10^{-10}$)	X
5 bp	0.1 ($<1 \cdot 10^{-320}$)	0.05($4.4 \cdot 10^{-121}$)	-0.003($2.4 \cdot 10^{-2}$)	X
6 bp	0.09($<1 \cdot 10^{-320}$)	0.02($8.9 \cdot 10^{-18}$)	$7.5 \cdot 10^{-4}$ (0.76)	X

Table S2 Poisson multiple regression coefficients and p-values for the effect on the mDNM rate by RRT length, repeat purity, GC content and motif length for all markers and stratified on motif length. Repeat tract length and repeat purity remain significant and consistent in their effect directionality across the full data set and all motif length subsets. GC-content is positively correlated to the mDNM rate for homopolymers but for di-, tri-, tetra-, and pentanucleotide repeats the motif GC-content has an inverse correlation to the mDNM rate.

Bp	1 bp motif	2 bp motif	3 bp motif	4 bp motif	5 bp motif	6 bp motif
11-20	0.13($3.2 \cdot 10^{-93}$)	0.04($1.2 \cdot 10^{-9}$)	0.05($5.7 \cdot 10^{-4}$)	-0.005(0.36)	-0.006(0.25)	-0.01(0.17)
21-30	0.21($< 1.0 \cdot 10^{-320}$)	0.16($< 1.0 \cdot 10^{-320}$)	0.13($3.4 \cdot 10^{-38}$)	0.05($2.0 \cdot 10^{-35}$)	-0.002(0.56)	0.002(0.65)
31-40	0.20($2.0 \cdot 10^{-36}$)	0.17($< 1.0 \cdot 10^{-320}$)	0.18($7.3 \cdot 10^{-136}$)	0.11($2.2 \cdot 10^{-211}$)	0.06($1.1 \cdot 10^{-28}$)	0.01(0.01)
41-50	0.06(0.23)	0.16($< 1.0 \cdot 10^{-320}$)	0.12($1.4 \cdot 10^{-95}$)	0.09($1.3 \cdot 10^{-293}$)	0.08($4.4 \cdot 10^{-40}$)	0.03($1.0 \cdot 10^{-6}$)
51-60	-0.50($3.7 \cdot 10^{-4}$)	0.15($3.0 \cdot 10^{-252}$)	0.11($5.1 \cdot 10^{-40}$)	0.07($5.8 \cdot 10^{-216}$)	0.06($3.5 \cdot 10^{-24}$)	0.06($1.1 \cdot 10^{-14}$)
61-70	-1.13($5.9 \cdot 10^{-5}$)	0.12($1.6 \cdot 10^{-36}$)	0.12($1.8 \cdot 10^{-12}$)	0.06($3.9 \cdot 10^{-115}$)	0.06($1.4 \cdot 10^{-18}$)	0.06($2.2 \cdot 10^{-9}$)
71-80	0.65(0.48)	0.14($9.3 \cdot 10^{-10}$)	0.03(0.34)	0.04($4.4 \cdot 10^{-42}$)	0.03($6.1 \cdot 10^{-6}$)	0.02(0.15)
81-90	X	0.05(0.58)	0.47(0.31)	0.04($7.6 \cdot 10^{-12}$)	0.05($5.0 \cdot 10^{-5}$)	0.06(0.06)
>90	X	X	5.43(1.00)	0.03($4.3 \cdot 10^{-5}$)	0.008(0.42)	0.04(0.34)

Table S3 Poisson regression coefficients and p-values for effect of repeat purity on mDNM rate split into ten bp RRT length bins stratified on motif length. Data for regression was not available for homopolymers above 80 bp and dinucleotide microsatellites above 90 bp.

Chrom	Pos	Motif	Ref. (No repeats)	Offspring gt	Father gt	Mother gt	Verified
chr11	80254522	CTAT	15	15/15	11/16	15/15	yes
chr12	100919737	TATC	15	15/18	15/17	15/17	yes
chr2	199703012	GTTT	8	9/9	8/10	8/9	yes
chr4	115864579	TG	11	11/13	11/12	11/11	no
chr7	86946160	CA	16	14/18	16/18	15/18	yes
chr11	43435069	AGAT	16	17/17	15/18	15/17	yes
chr13	54157465	TA	20	16/19	18/21	16/22	yes
chr14	82746964	AATA	8	8/10	8/9	8/9	yes
chr17	54067449	TTTTA	7	7/8	7/7	7/7	yes
chr18	3742950	CTTC	11	11/14	11/12	11/13	yes
chr18	63468251	AAC	8	10/12	8/10	10/10	yes
chr1	10498281	AAAAC	8	8/7	9/9	7/9	yes
chr4	140179641	TGTTT	8	7/9	8/5	5/7	yes
chr6	44850746	ATCT	13	16/16	15/15	10/16	yes
chr16	69147179	AAT	15	14/16	15/15	15/14	yes
chr1	180853627	AC	21	19/25	19/26	19/20	yes
chr20	34877898	AGAT	12	15/16	13/16	16/16	yes
chr3	85784531	TATC	13	13/12	10/14	11/12	yes
chr4	169443052	TATC	13	12/15	14/14	13/12	yes
chr6	38464486	TTTTC	6	6/5	6/7	6/7	yes
chr7	29184144	AGAT	14	14/15	11/16	14/13	yes
chr8	110980554	ATCT	10	12/12	11/11	12/13	yes
chr11	22272303	CA	22	22/19	23/26	22/25	yes
chr12	29413662	AC	15	16/20	14/16	14/19	yes
chr13	80326450	TG	9	9/8	8/8	8/8	yes
chr1	172791800	AT	7	7/14	7/7	7/13	yes
chr8	19119757	TA	9	9/13	9/12	9/12	yes

Table S4 mDNMs verified by haplotype resolved assemblies generated from pacBio HiFi sequencing data. We were unable to verify homopolymer mDNMs due to high error rates.

Marker 1	Marker 2	Conditional *P* 1-2	Conditional *P* 2-1	R^2
chr2:47416318	chr2:47494068	$8.9 \cdot 10^{-2}$	$5.1 \cdot 10^{-2}$	0.74
chr2:47416318	chr2:47491330:0	$3.6 \cdot 10^{-2}$	$2.3 \cdot 10^{-2}$	0.60
chr2:47416318	chr2:47491330:1	$3.6 \cdot 10^{-2}$	$2.3 \cdot 10^{-2}$	0.60
chr2:47494068	chr2:47491330:0	0.17	0.18	0.80
chr2:47494068	chr2:47491330:1	0.17	0.18	0.80
chr2:47491330:0	chr2:47491330:1	1.00	1.00	1.00

Table S5 Conditional association p-values for markers significantly associating with increased mDNM rate (*MSH2* missense marker in bold). All markers are correlated but a residual signal remains after conditioning the intergenic signal at chr2:47491330 for the missense marker

Population	Allele frequency of rs4987188
Iceland	1.9%
UKB	1.5%
Fingen	3.0%
Africa	0.2%
Asia	0.0.%

Table S6 Allele frequency of rs4987188 in different populations, it is most frequent in the Finnish population and not found in Asians.

Motif (bp)	Paternal	Motifs affected	Maternal	Motifs affected	Mann-Whitney *P*
1	5,141(14.5%)	1.67	2,962(27.6%)	2.13	**$1.9 \cdot 10^{-23}$**
2	17,388(49.0%)	1.38	4,119(38.4%)	1.71	**$6.0 \cdot 10^{-13}$**
3	2,663(7.5%)	1.22	887(8.2%)	1.33	$5.5 \cdot 10^{-2}$
4	9,184(25.9%)	1.06	2,365(22.0%)	1.12	**$9.6 \cdot 10^{-5}$**
5	977(2.8%)	1.08	343(3.2%)	1.21	0.38
6	148(0.4%)	0.95	62(0.6%)	0.77	0.26

Table S7 Motif length composition of paternal and maternal mDNMs, mean number of motifs added/removed for each motif length in each parent and Mann-Whitney U test p-values for different step sizes between parents. Bold represents significant difference in step size between parents ($p<0.05$).

Motif length (bp)	Mean RRT length(bp) (pat/mat)	*P*
1	17.0/16.9	0.12
2	39.2/37.2	**$1.9 \cdot 10^{-24}$**
3	40.8/40.4	0.72
4	52.2/50.1	**$5.6 \cdot 10^{-9}$**
5	54.7/51.1	**$1.6 \cdot 10^{-3}$**
6	49.8/42.6	**$2.2 \cdot 10^{-2}$**

Table S8 Mean RRT lengths for paternal and maternal mDNMs. The paternal mDNMs have longer RRT at di-, tetra-, penta- and hexanucleotide microsatellites. Bold represents significant difference in RRT length between paternal and maternal mDNMs ($p<0.05$).

Members of ATTCT motif equivalence class
ATTCT
TTCTA
TCTAT
CTATT
TATTC
AGAAT
GAATA
AATAG
ATAGA
TAGAA

Table S9 Members of the ATTCT motif equivalence class considered for paternal expansion bias.

Motif equivalence class	% of paternal	% of maternal	Odds ratio maternal/paternal (*P*)
AAC	10.7%	7.6%	**0.68[0.52;0.90] ($7.6 \cdot 10^{-3}$)**
AAG	2.1%	2.7%	1.28[0.77;2.05] (0.36)
AAT	80.9%	84.2%	**1.26[1.03;1.55] (0.02)**
ACC	0.3%	0.0%	No maternal mDNMs.
ACG	0.0%	0.0%	No mDNMs
ACT	1.6%	0.8%	0.49[0.20;1.04] ($7.2 \cdot 10^{-2}$)
AGC	0.6%	1.4%	2.14[0.99;4.50] ($5.1 \cdot 10^{-2}$)
AGG	0.2%	0.2%	1.56[0.19;8.48] (0.64)
ATC	3.4%	2.1%	0.63[0.37;1.02] ($7.2 \cdot 10^{-2}$)
CCG	0.3%	1.0%	**3.39[1.28;9.20] ($2.0 \cdot 10^{-2}$)**
Motif equivalence class	% of paternal	% of maternal	Enrichment
AC	88.7%	79.8%	**0.50[0.46;0.55] ($2.9 \cdot 10^{-48}$)**
AG	4.6%	7.4%	**1.65[1.43;1.88] ($4.5 \cdot 10^{-12}$)**
AT	6.6%	12.7%	**2.05[1.84;2.29] ($3.6 \cdot 10^{-35}$)**
CG	0.04%	0.1%	3.04[0.88;9.76] ($6.2 \cdot 10^{-2}$)

Table S10 Motif equivalence class odds ratios for di- and trinucleotide microsatellites. Maternal mDNMs are more common at AAT, CCG, AG and AT motif class microsatellites while paternal mDNMs are more common at AAC and AC motif class microsatellites. Bold represents significant enrichment ($p<0.05$).

	Coefficient (*P*) for lower step size than homopolymers	
Motif length	Paternal	Maternal
2	$-0.3(5.3 \cdot 10^{-50})$	$-0.5(2.0 \cdot 10^{-22})$
3	$-0.4(3.2 \cdot 10^{-31})$	$-0.6(6.6 \cdot 10^{-15})$
4	$-0.5(7.6 \cdot 10^{-104})$	$-0.8(8.9 \cdot 10^{-41})$
5	$-0.5(5.4 \cdot 10^{-23})$	$-0.6(7.5 \cdot 10^{-7})$
6	$-0.6(7.8 \cdot 10^{-7})$	$-0.9(8.9 \cdot 10^{-4})$

Table S11 Linear regression coefficients and p-values for the effect of increasing motif length on step size in paternal and maternal mDNMs. All motif lengths have lower step sizes than homopolymers.

		Read pair tracing		
		Paternal	Maternal	Total
3 generation	Paternal	2,199	70	2,269(75.5%)
	Maternal	118	619	737(24.5%)
	Total	2,317(77.1%)	689(22.9%)	

Table S12 Comparison of phasing results for mDNMs phased using both three generation and read pair phasing. The concordance between the methods is 93.7% and the ratio between maternal and paternal mDNMs is similar in both sets.

		Allele based		
		Paternal	Maternal	Total
3 generation	Paternal	2,390	39	2,429(78.2%)
	Maternal	103	576	679(21.8%)
	Total	2,493(80.2%)	615(19.8%)	

Table S13 Comparison of phasing results for mDNMs phased using both three generation and allele based phasing. The concordance between the methods is 95.4% and the ratio between maternal and paternal mDNMs is similar in both sets.

		Read pair tracing		
		Paternal	Maternal	Total
Allele based	Paternal	8,865	9	8,874(79.6%)
	Maternal	21	2,258	2,279(20.4%)
	Total	8,886 (79.7%)	2,267(20.3%)	

Table S14 Comparison of phasing results for mDNMs phased using both read pair and allele based phasing. The concordance between the methods is 99.7%.

Motif length (bp)	Transmission rate	Error estimate
1	0.40 (861/2,156)	20%
2	0.50 (2,558/5,083)	0%
3	0.51 (433/857)	2%
4	0.53 (1,458/2,764)	6%
5	0.55 (169/310)	10%
6	0.47 (27/58)	6%
Total	0.49 (5,506/11,228)	2%

Table S15 Transmission rate by motif length and combined for all mDNMs phased using haplotype sharing in three-generation families.

Motif class	Maternal	Paternal	Full marker set	TRF output
A	97.5%	97.7%	99.2%	99.2%
C	2.5%	2.3%	0.8%	0.8%
AC	67.6%	84.3%	49.1%	54.4%
AG	8.9%	5.6%	19.9%	23.4%
AT	23.2%	10.0%	30.5%	21.7%
CG	0.4%	0.1%	0.4%	0.4%
AAC	13.0%	13.3%	27.1%	20.8%
AAG	3.8%	2.0%	15.0%	10.0%
AAT	75.3%	77.6%	34.9%	32.9%
ACC	0.8%	0.4%	4.0%	7.4%
ACG	0.0%	0.0%	0.1%	0.1%
ACT	1.2%	0.9%	0.9%	1.9%
AGC	2.1%	1.2%	3.0%	7.8%
AGG	0.8%	0.6%	7.5%	11.3%
ATC	2.4%	3.9%	4.3%	6.1%
CCG	0.6%	0.2%	3.1%	1.6%

Table S16 Fraction of each motif equivalence class within homopolymers, di- and trinucleotide repeats split on maternal and paternal mDNMs, fraction of each motif equivalence class in full marker set and in the Tandem Repeats Finder (TRF) output.

Motif length (bp)	All microsatellites	Microsatellites in coding exons	Non polymorphic STRs in coding exons
1	50.0%	1.8%	0.003%
2	16.7%	0.5%	0.6%
3	5.8%	76.2%	17.5%
4	13.2%	3.0%	6.4%
5	7.1%	1.5%	22.1%
6	6.5%	17.1%	53.4%

Table S17 Motif length composition for all microsatellites and exon intersecting microsatellites.

Motif length	All microsatellites	Coding microsatellites	Non-polymorphic STRs	Coding non-polymorphic STRs
1	0.98	0.97 (0.25)	0.97	0.95 (0.05)
2	0.95	0.96 (0.14)	0.93	0.97 ($5.9 \cdot 10^{-12}$)
3	0.93	0.89 ($9.0 \cdot 10^{-66}$)	0.94	0.93 ($1.3 \cdot 10^{-8}$)
4	0.89	0.88 (0.87)	0.93	0.96 ($5.7 \cdot 10^{-43}$)
5	0.84	0.89 ($3.0 \cdot 10^{-4}$)	0.96	0.97 ($2.3 \cdot 10^{-26}$)
6	0.81	0.80 (0.12)	0.94	0.94 ($3.8 \cdot 10^{-4}$)
Overall	0.94	0.87 ($1 \cdot 10^{-153}$)	0.95	0.95 ($9.6 \cdot 10^{-14}$)

Table S18 Average purity values per motif length for all microsatellites, coding microsatellites, all non-polymorphic STRs and coding non-polymorphic STRs. Mann-Whitney U-test p-values for significant difference in purity values between exonic and non-exonic microsatellites in each motif length in brackets.

RRT length	Purity effect (Ice/UKBB)	*P* (Ice/UKBB)
11-20	0.39/0.23	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
21-30	0.71/0.67	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
31-40	0.87/0.85	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
41-50	1.17/1.15	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
51-60	1.17/1.18	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
61-70	1.10/1.13	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
71-80	1.05/1.08	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
81-90	0.96/1.07	$3.9 \cdot 10^{-141}$/$6.8 \cdot 10^{-183}$
>90	0.76/0.83	$5.4 \cdot 10^{-56}$/$6.1 \cdot 10^{-57}$

Table S19 Linear regression coefficients and p-values for the effect of repeat purity on expected heterozygosity within ten base pair bins of RRT length for both the Icelandic and UKB datasets.

RRT length	Purity effect (Ice/UKBB)	*P* (Ice/UKBB)
11-20	7.86/5.31	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
21-30	7.84/6.09	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
31-40	6.11/5.47	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
41-50	5.94/4.96	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
51-60	4.78/3.668	$< 1.0 \cdot 10^{-320}$/$2.4 \cdot 10^{-83}$
61-70	4.42/2.72	$2.8 \cdot 10^{-164}$/$6.0 \cdot 10^{-17}$
71-80	4.22/1.95	$2.0 \cdot 10^{-61}$/$7.1 \cdot 10^{-6}$
81-90	4.03/-0.09	$6.2 \cdot 10^{-26}$/0.9
>90	2.72/0.61	$1.0 \cdot 10^{-16}$/0.1

Table S20 Logistic regression coefficients and p-values for the effect of repeat purity on polymorphism rate within ten base pair bins of RRT length for both the Icelandic and UKB datasets.

Motif length	Effect (Ice/UKBB)	*P* (Ice/UKBB)
1	0.15/0.17	$4.5 \cdot 10^{-312}$/$< 1.0 \cdot 10^{-320}$
2	0.14/0.10	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
3	-0.12/-0.05	$< 1.0 \cdot 10^{-320}$/$< 1.0 \cdot 10^{-320}$
4	-0.07/-0.04	$1.1 \cdot 10^{-155}$/$< 1.0 \cdot 10^{-320}$
5	-0.06/-0.03	$8.0 \cdot 10^{-141}$/$< 1.0 \cdot 10^{-320}$
6	-0.02/-0.01	$3.9 \cdot 10^{-18}$/$1.52 \cdot 10^{-95}$

Table S21 Linear regression coefficients and p-values for the effect of GC motif content on expected heterozygosity within motif lengths for both the Icelandic and UKB datasets.

Motif length	Effect (Ice/UKBB)	*P* (Ice/UKBB)
1	-0.20/-1.33	$4.9 \cdot 10^{-3}/4.6 \cdot 10^{-41}$
2	-0.29/-0.21	$1.3 \cdot 10^{-79}/1.6 \cdot 10^{-15}$
3	-0.79/-1.12	$< 1.0 \cdot 10^{-320}/< 1.0 \cdot 10^{-320}$
4	-1.47/-1.97	$< 1.0 \cdot 10^{-320}/< 1.0 \cdot 10^{-320}$
5	-2.02/-1.67	$< 1.0 \cdot 10^{-320}/< 1.0 \cdot 10^{-320}$
6	-1.55/-1.34	$< 1.0 \cdot 10^{-320}/< 1.0 \cdot 10^{-320}$

Table S22 Logistic regression coefficients and p-values for the effect of GC motif content on polymorphism rate within motif lengths for both the Icelandic and UKB datasets.

Motif equivalence class	Regression α and (*P*) for comparison with AC-class mDNM rate
AG	-1.20 ($<1 \cdot 10^{-320}$)
AT	-1.08 ($<1 \cdot 10^{-320}$)
CG	-0.43 ($7.7 \cdot 10^{-3}$)

Table S23 Regression coefficients and p-values from a Poisson regression comparing the mDNM rate of the other motif equivalence classes to the AC class using available trios per marker as exposure but without correcting for RRT length. All motif classes have significantly lower mDNM rates.

Motif equivalence class	Regression α and (*P*) for comparison with CG-class mDNM rate
AC	-0.81 ($5.7 \cdot 10^{-7}$)
AG	-1.48 ($1.7 \cdot 10^{-19}$)
AT	-1.18 ($5.8 \cdot 10^{-13}$)

Table S24 Regression coefficients and p-values from a Poisson regression comparing the mDNM rate of the other motif equivalence classes to the CG class after correcting for RRT length and using available trios per marker as exposure. All motif classes have significantly lower mDNM rates.

Motif equivalence class	Regression α (*P*) for comparison with AAT mDNM rate
AAC	-1.35 ($2.7 \cdot 10^{-217}$)
AAG	-1.67 ($5.3 \cdot 10^{-75}$)
ACC	-2.99 ($5.6 \cdot 10^{-29}$)
ACG	-0.06 (0.92)
ACT	-2.49 ($3.3 \cdot 10^{-88}$)
AGC	-1.8 ($8.6 \cdot 10^{-41}$)
AGG	-3.09($4.9 \cdot 10^{-37}$)
ATC	-1.78 ($1.2 \cdot 10^{-120}$)
CCG	-1.14($1.7 \cdot 10^{-11}$)

Table S25 Regression coefficients and p-values from a Poisson regression comparing the mDNM rate of the other motif equivalence classes to the AAT class, correcting for RRT length and using available trios per marker as exposure. All motif classes expect for the rarest one (ACG) have significantly lower mDNM rates.

Motif length (bp)	Odds ratio maternal/paternal (95% CI)	Fisher exact test *P*
1	2.25 (2.14-2.37)	$2.0 \cdot 10^{-198}$
2	0.65 (0.62-0.68)	$6.3 \cdot 10^{-84}$
3	1.11 (1.03-1.20)	$1.0 \cdot 10^{-2}$
4	0.81 (0.77-0.85)	$4.2 \cdot 10^{-16}$
5	1.17 (1.03-1.32)	$1.7 \cdot 10^{-2}$
6	1.39 (1.03-1.86)	$3.0 \cdot 10^{-2}$
Motif length (without 1bp)	Odds ratio maternal/paternal (95% CI) (no 1 bp)	Fisher exact test *P* (no 1 bp)
2	0.84 (0.80-0.88)	$1.0 \cdot 10^{-11}$
3	1.34 (1.24-1.45)	$3.3 \cdot 10^{-12}$
4	1.01 (0.95-1.06)	0.78
5	1.39 (1.22-1.57)	$6.3 \cdot 10^{-7}$
6	1.64 (1.21-2.20)	$1.9 \cdot 10^{-3}$

Table S26 Per motif length enrichment of mDNMs between maternally and paternally phased mDNMs. Top half shows enrichment including homopolymers and bottom half without them. Tri-, penta- and hexanucleotide repeats are enriched in maternal mDNMs while dinucleotide microsatellites are paternally enriched.

Length threshold (bp)	percentage adding bp below	percentage adding bp above
20	59.4%	52.1%
30	59.6%	50.7%
40	56.9%	49.5%
50	54.2%	49.9%
60	53.5%	51.2%
70	53.5%	50.2%
80	53.4%	46.2%
90	53.4%	50.0%
100	53.4%	42.9%

Table S27 Fraction of mDNMs adding bp above and below different repeat tract length thresholds. The fraction of mDNMs adding bp is higher below the threshold in all cases.

Detection frequency	DNM rate estimate
$9.0 \cdot 10^{-6}$	$5.0 \cdot 10^{-5}$
$7.6 \cdot 10^{-5}$	$5.6 \cdot 10^{-5}$
$1.0 \cdot 10^{-2}$	$8.1 \cdot 10^{-5}$
$5.0 \cdot 10^{-2}$	$9.3 \cdot 10^{-5}$
0.1	$1.0 \cdot 10^{-4}$

Table S28 Microsatellite mDNM rate estimates for different detection frequencies, the number of markers included in the analysis depends on the detection frequency defined by the sample size.

Phasing method	Paternal effect	*P*	Maternal effect	*P*
Allele based	0.17 [0.15;0.18]	$1.2 \cdot 10^{-102}$	0.07 [0.05;0.08]	$5.1 \cdot 10^{-16}$
Read back	0.16 [0.15;0.18]	$3.6 \cdot 10^{-101}$	0.07 [0.06;0.09]	$7.1 \cdot 10^{-21}$
3 generation	0.18 [0.15;0.21]	$9.8 \cdot 10^{-34}$	0.05 [0.03;0.07]	$1.7 \cdot 10^{-6}$

Table S29 Poisson regression coefficients and p-values for subsets of phased markers by each method applied. The effects for allele based phasing and read back phasing match but the three generation phasing subset gives a stronger paternal effect and a weaker maternal one.

Motif length (bp)	Maternal	Paternal	Maternal percentage
1	2,962	5,141	36.6%
2	4,119	17,388	19.2%
3	887	2,663	25.0%
4	2,365	9,184	20.5%
5	343	977	26.0%
6	62	148	29.5%
Total	10,738	35,501	23.2%

Table S30 Parental ratios of mDNMs stratified on motif length and on the full set.

Motif length (bp)	Shared	Not shared	Error rate
1	47	6	11.3%
2	108	6	5.3%
3	16	1	5.9%
4	44	0	0%
5	2	0	0%
6	0	0	0%

Table S31 Sharing of mDNMs between monozygotic twins stratified on motif length.

Author Contributions

Paper was written by SK, supervised by BVH, with input from HJ, OAS, GAA, UT, TR, DFG and KS. SK, HPE, AG, PIO, GM and BVH developed analysis tools, with input from HJ, OE and DFG. GP performed mDNM enrichment analysis. AG and DB generated SV genotypes. GS performed gene burden test. GH assembled and haplotype tagged PacBio HiFi data. Association analysis was performed by SK, TR and BVH. Age effect regression model was implemented by HJ. SNP mutation signatures were created by SO. SNP genotyping was performed by HPE and PIO. Microsatellite genotyping was performed by SK. Figures were drawn by MTH and SK. Study was supervised by BVH and KS. All authors agreed to the final version of the manuscript.